

Session 1: Microbiology to molecular biology 'Germs to Genes'

Molecular Approaches to Clinical Microbiology in Africa 2025

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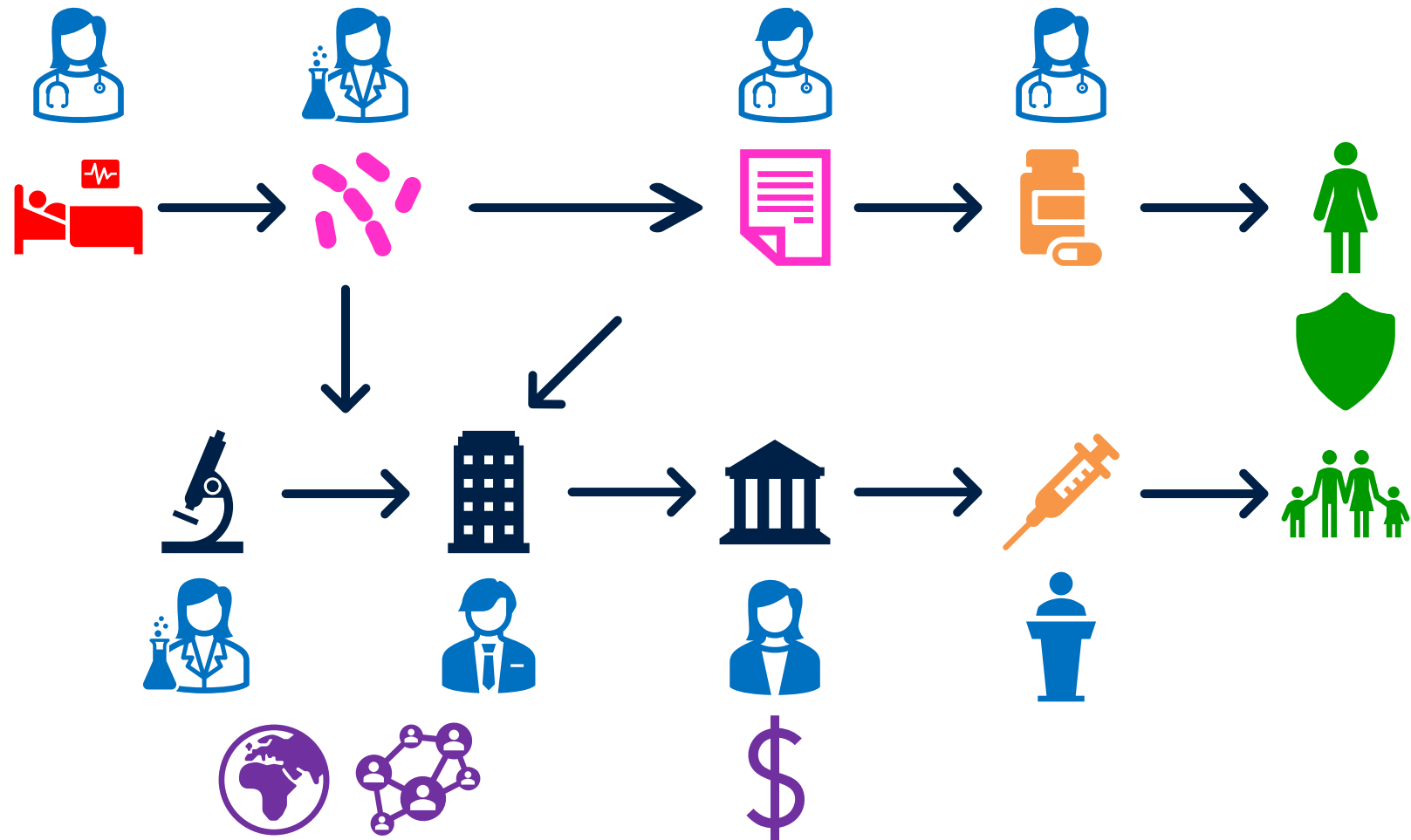


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OXFORD

Learning Outcomes

1. Overview of culture-based clinical microbiology.
2. The impact of molecular approaches.
3. Barriers to understanding molecular and genomic data.
4. How molecular data has affected our understanding of an exemplar disease, cholera.
5. 16S rRNA as an exemplar molecular approach, including an online exercise.

Clinical and public health microbiology without molecular approaches

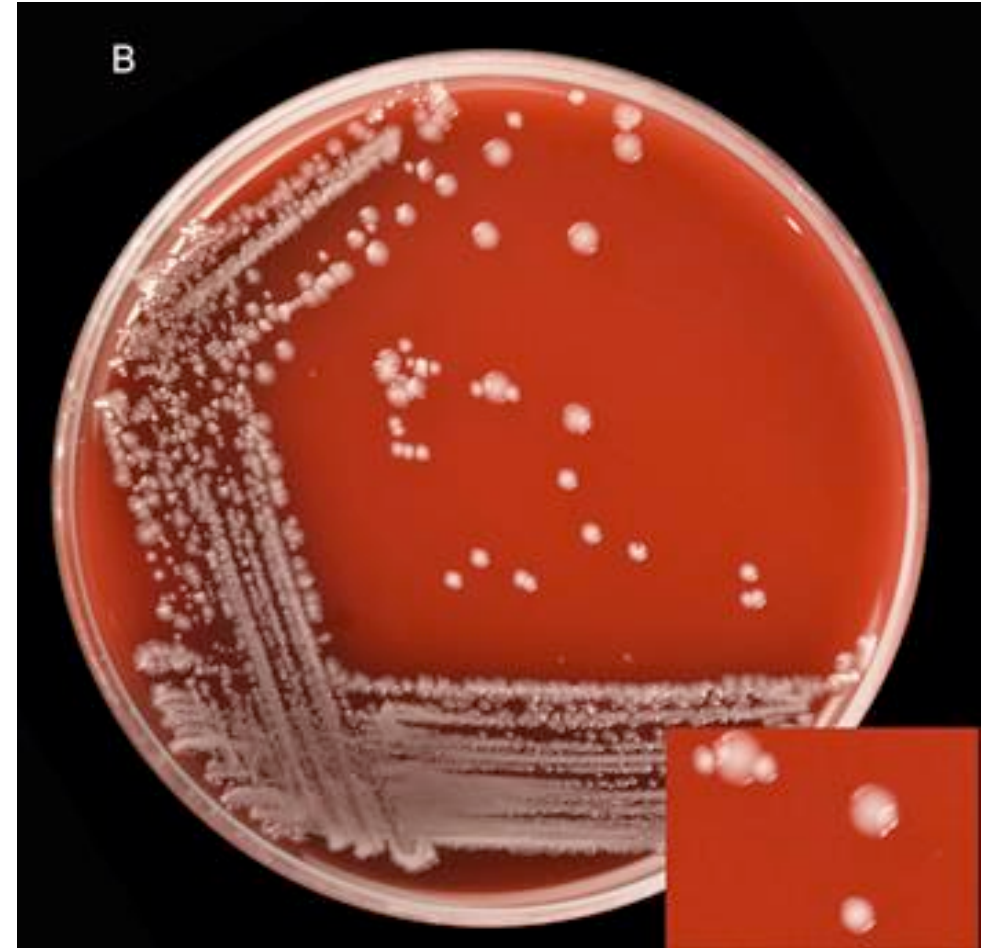


Microbiology: pure culture

- The ability to isolate pathogens was foundational to microbiology.
 - Koch's postulates.
- In addition to diagnosis, this provides material for further characterisation.
 - Isolates remain widely used.

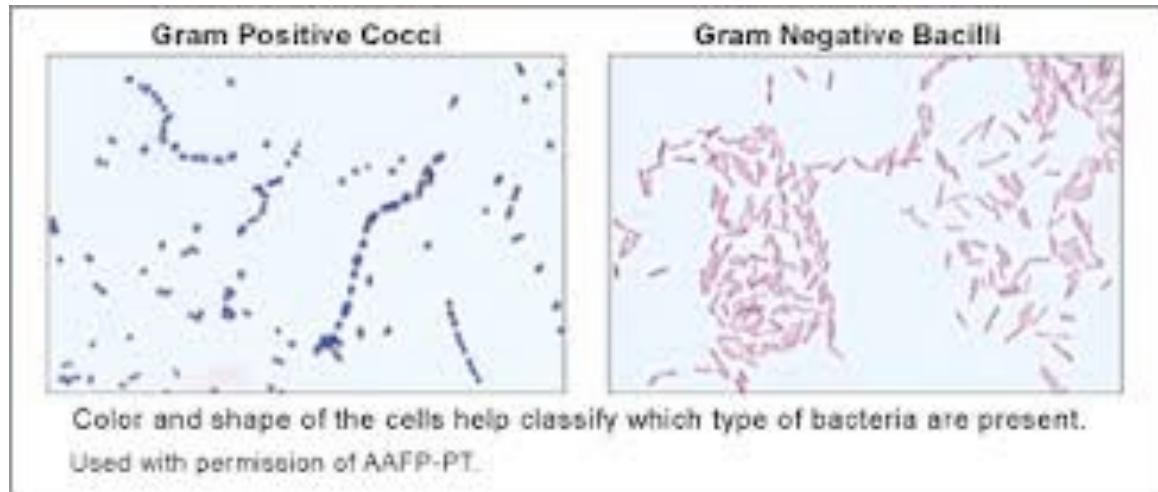
However:

- culture requires sophisticated, often bespoke, reagents and equipment, and time;
- many pathogens cannot be grown cultured:
 - lack of methods;
 - lack of suitable specimens.

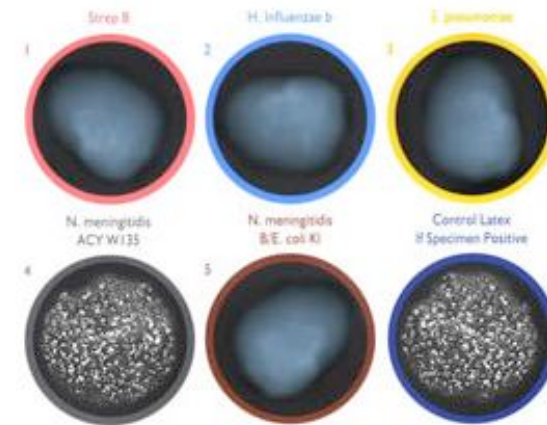


Phenotypic characterisation

Gram's Stain



Serology

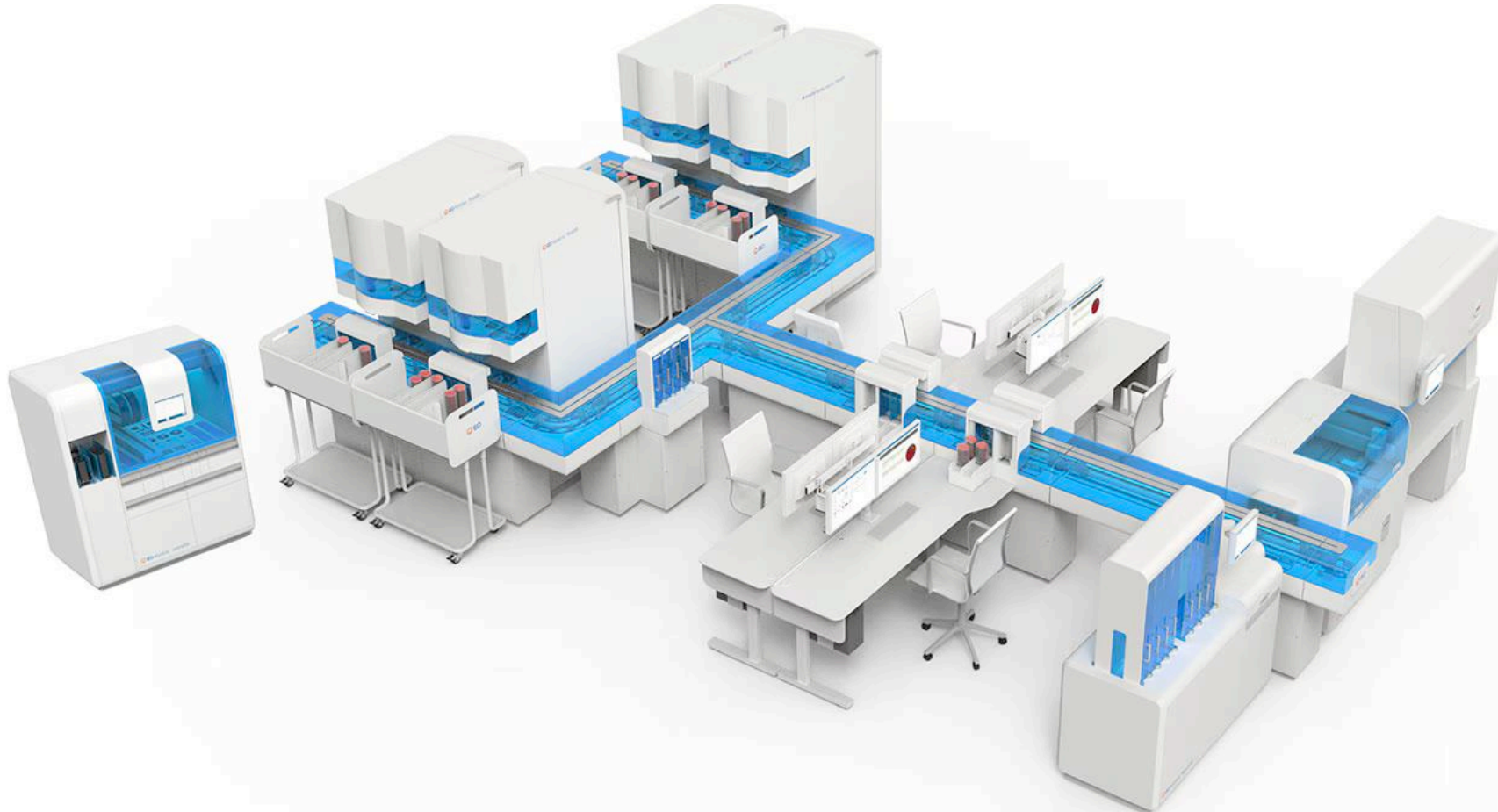


Anti-microbial
resistance
testing

Metabolic
phenotyping
For species
identification

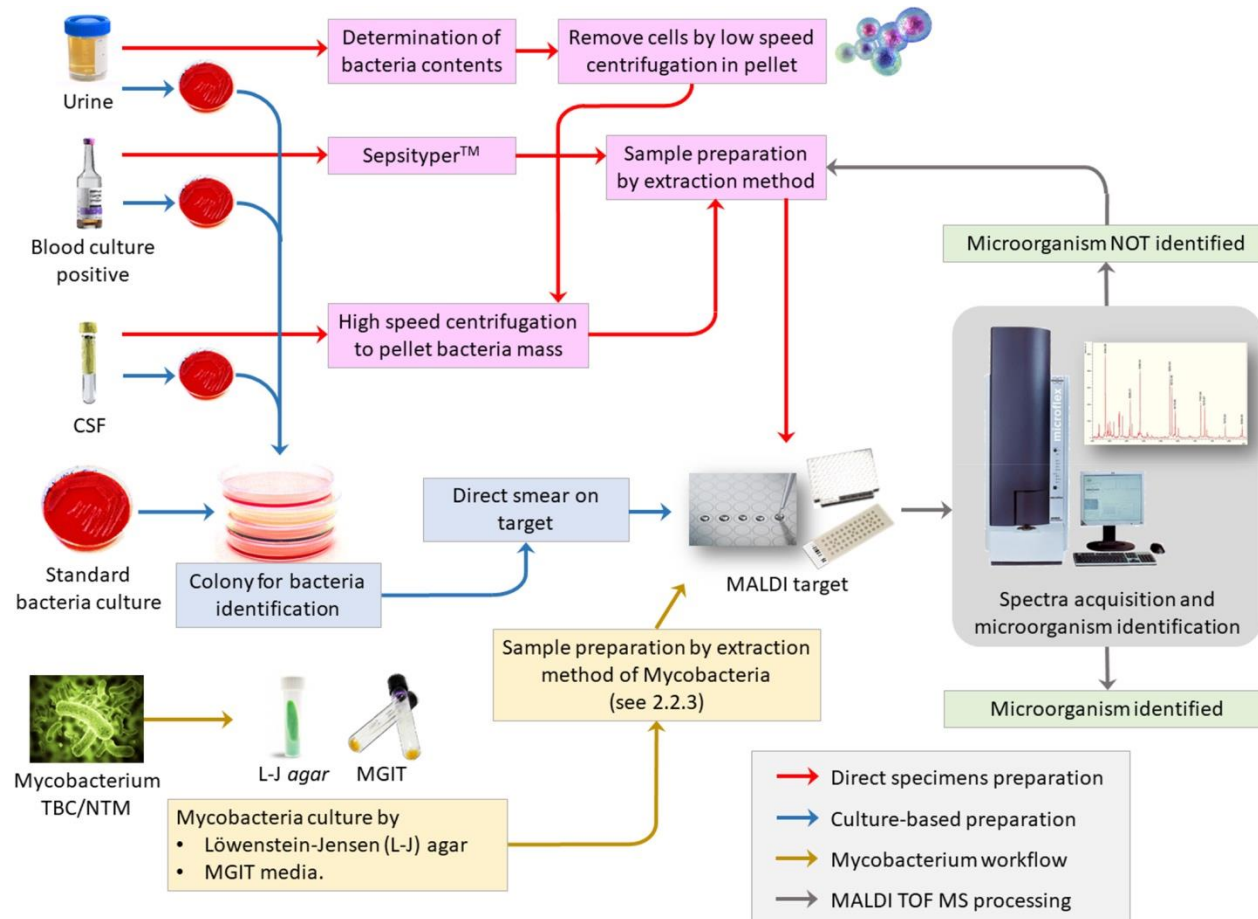


Automating culture



<https://www.bd.com/en-us/products-and-solutions/products/product-brands/kiestra>

MALDI-TOF phenotyping



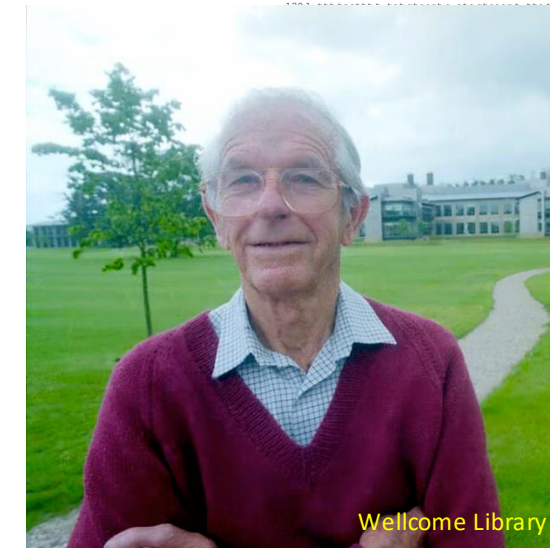
Matrix-assisted laser desorption/ionization-time of flight (MALDI-TOF) mass spectrometry.

- Enables rapid and inexpensive (per sample) species identification
- Requires:
 - bacterial sample (usually an isolate);
 - the equipment;
 - a database linking spectra to bacterial species identification.
- Commercial systems are widely installed and used in clinical laboratories.

Hou, T. Y., Chiang-Ni, C. & Teng, S. H. (2019). Current status of MALDI-TOF mass spectrometry in clinical microbiology. *J Food Drug Anal.* **27**, 404-414.

The molecular revolution and DNA sequences

- **Definitive:**
 - fundamental level of information;
 - any part of the genome can be accessed.
- **Reproducible:**
 - nucleotide sequences are either right or wrong and can be checked;
 - reverse mutations are (usually) rare.
- **Scalable:**
 - nucleotide sequencing technology can be conducted on one or many samples and on a few base pairs or a whole genome.
- **Manipulable:**
 - nucleotide sequences can be analysed with model-based methods.
- **Can be done from a PCR reaction or microbiome sample:**
 - an isolate is not necessarily required.



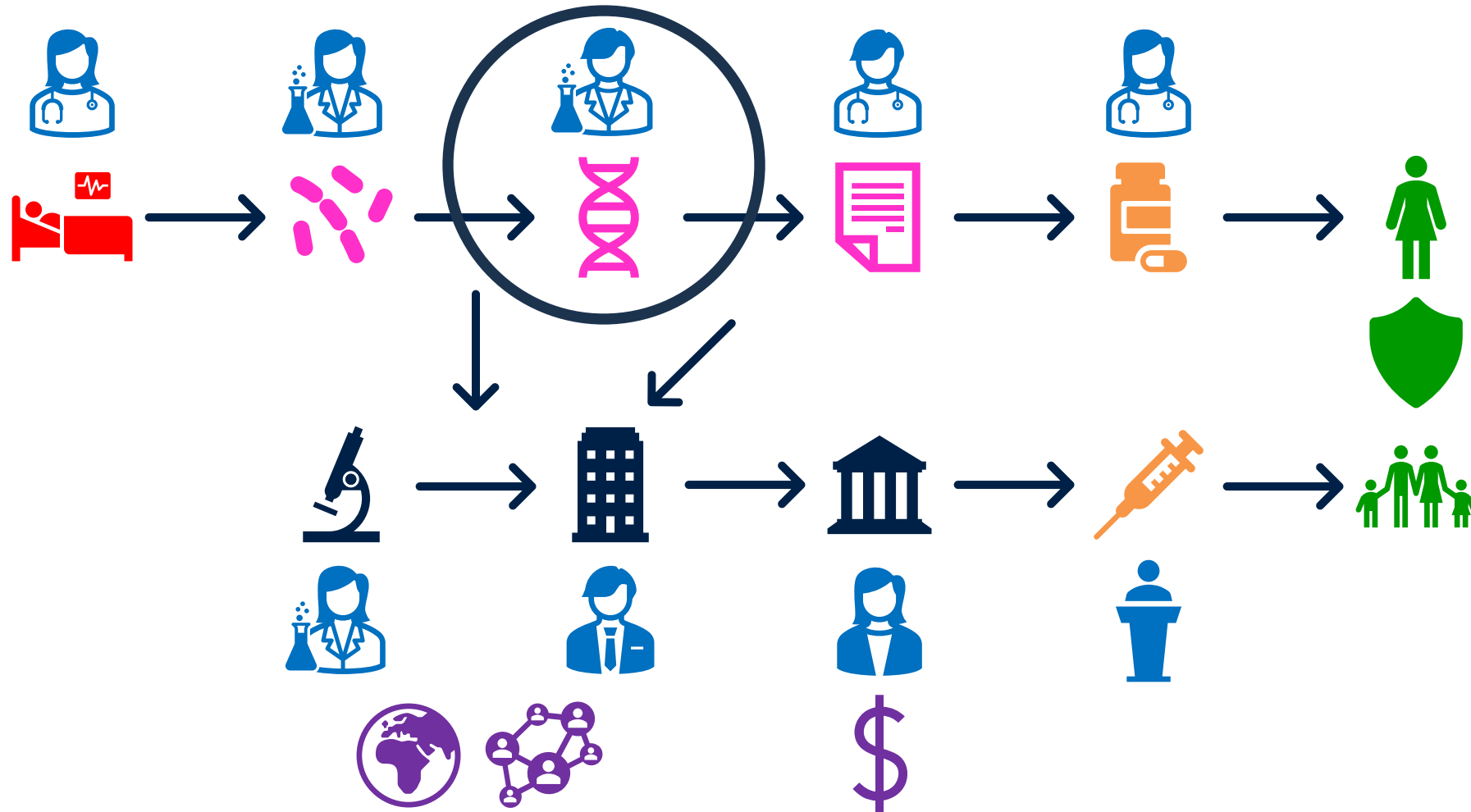
Wellcome Library

Frederick Sanger (1918-2013)

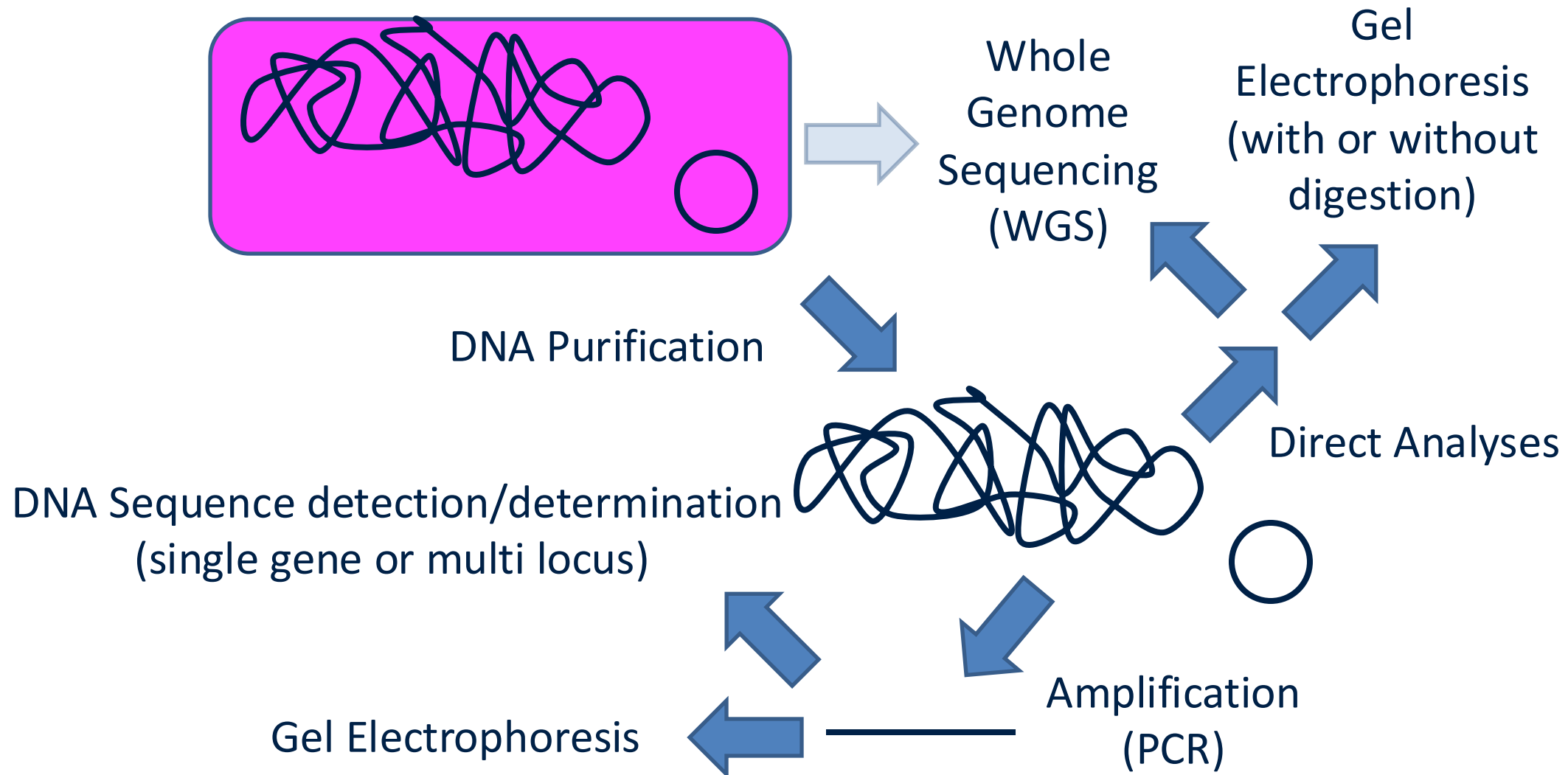
Sanger, F., Air, G. M., Barrell, B. G., Brown, N. L., Coulson, A. R., Fiddes, C. A., Hutchison, C. A., Slocombe, P. M. & Smith, M. (1977). Nucleotide sequence of bacteriophage phi X174 DNA. *Nature* 265, 687-695.

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Molecular approaches to clinical and public health microbiology



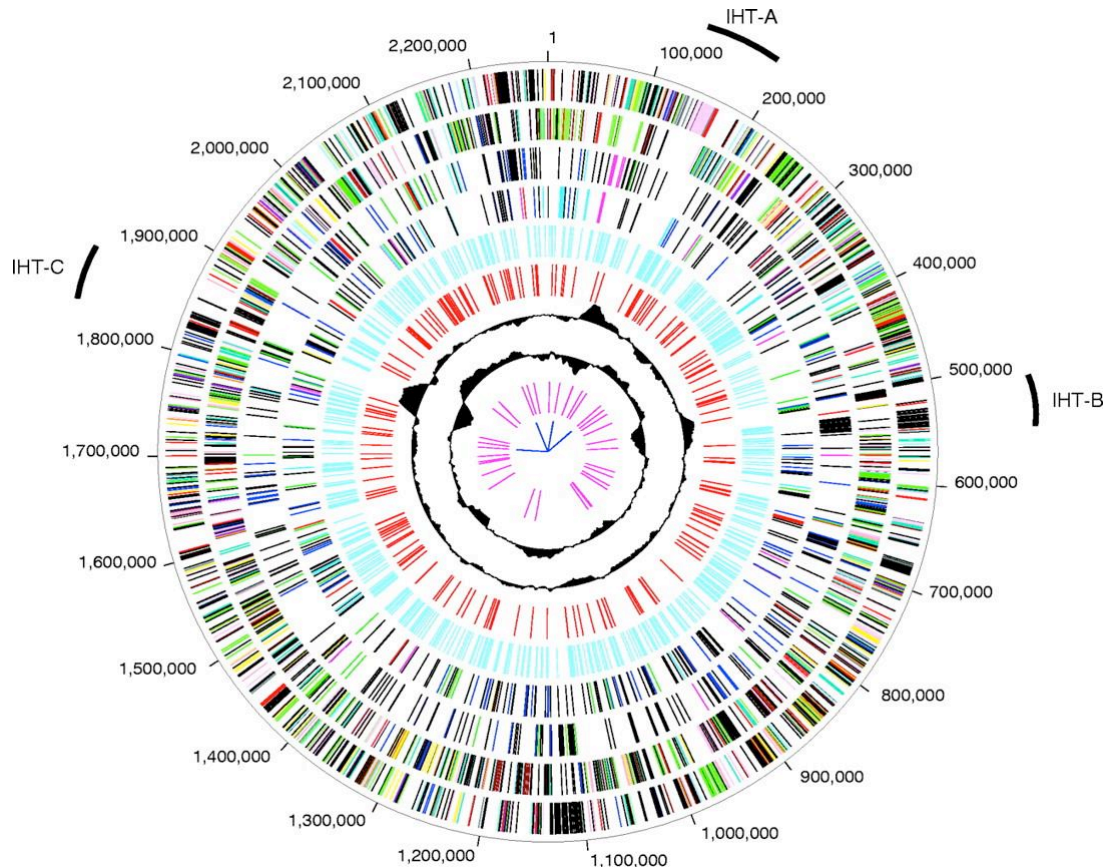
Microbial DNA and what we do with it



The role of genetics and genomics

- Nucleotide sequencing facilitates specimen characterisation,
 - sequence-based typing, up to the level of whole genome sequences (WGSs).
- Sequences are definitive, reproducible, and comparable.
 - easily stored, transported, and manipulated with analysis algorithms;
- However, datasets are large and complex,
 - requiring interpretation for the practitioner.

Translating molecular and genomic data



Barriers:

- unfamiliarity;
- complexity, apparent and real;
- applicability.

Solutions:

- explaining & summarising;
- structuring & curating;
- interpreting & synthesising.

Tettelin, H., Saunders, N. J., Heidelberg, J., Jeffries, A. C., Nelson, K. E., Eisen, J. A., Ketchum, K. A., Hood, D. W., Peden, J. F., Dodson, R. J., Nelson, W. C., Gwinn, M. L., DeBoy, R., Peterson, J. D., Hickey, E. K., Haft, D. H., Salzberg, S. L., White, O., Fleischmann, R. D., Dougherty, B. A., Mason, T., Ciecko, A., Parksey, D. S., Blair, E., Cittone, H., Clark, E. B., Cotton, M. D., Utterback, T. R., Khouri, H., Qin, H., Vamathevan, J., Gill, J., Scarlato, V., Massignani, V., Pizza, M., Grandi, G., Sun, L., Smith, H. O., Fraser, C. M., Moxon, E. R., Rappuoli, R. & Venter, J. C. (2000). Complete genome sequence of *Neisseria meningitidis* serogroup B strain MC58. *Science* **287**, 1809-1815.

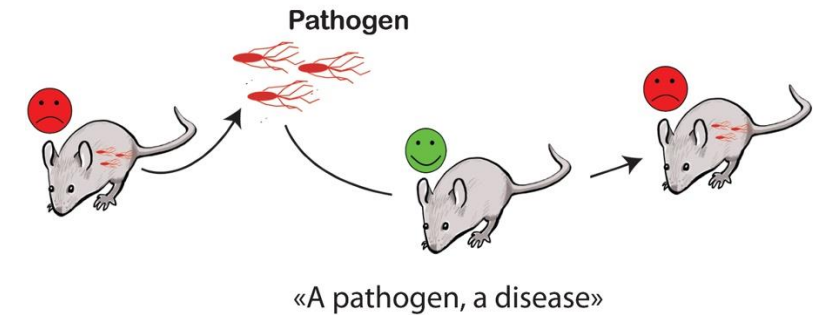
Impacts of molecular data on clinical microbiology principles and practice

Koch's Postulates revisited

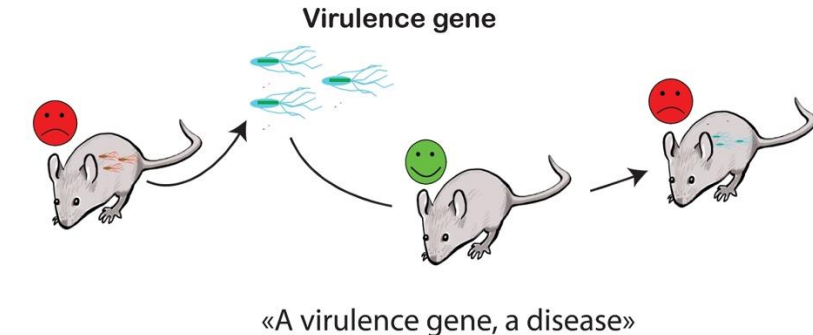
- Koch's postulates original formulation is, the:
 - microbe is found in all cases of disease and is absent in its absence;
 - microbe can be isolated and grown in pure culture;
 - cultured microbe can cause disease in a healthy host;
 - microbe can be re-isolated and cultured from this host.
- These ideas have been revisited in the light of knowledge on molecular pathology and the microbiome.
 - 'Molecular Koch's Postulates';
 - 'Ecological Koch's Postulates'.

Vonaesch, P., Anderson, M. & Sansonetti, P. J. (2018). Pathogens, microbiome and the host: emergence of the ecological Koch's postulates. *FEMS Microbiol Rev.* **42**, 273-292.

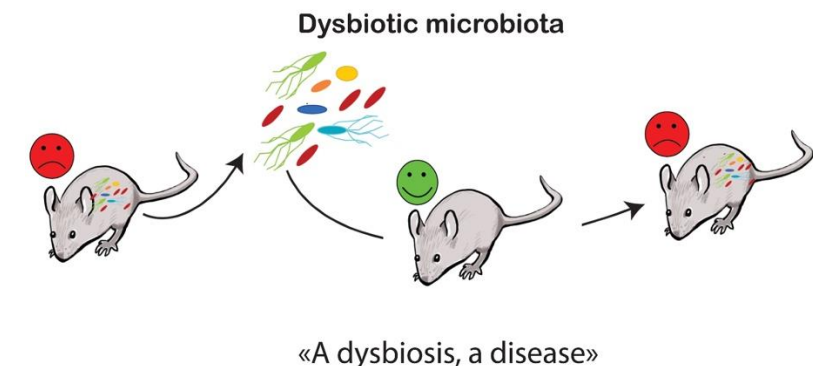
Koch's postulate



Molecular Koch's postulate



Ecological Koch's postulate



Case study: Cholera



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Climate change News feature 8 April 2024

Why is cholera killing thousands in southern Africa?

'If we were doing enough, we would not have seen such a massive cholera outbreak.'



Freddie Clayton
British journalist who covers climate, the environment, and public health



Kennedy Phiri
A Zambian journalist reporting on environmental and gender issues, and the co-founder of Media Network Action on Climate Change.



Margaret Courtney Clarke

Southern Africa's informal settlements often lack adequate water, sanitation, and hygiene infrastructure, like this informal settlement in Windhoek, Namibia.

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<https://www.thenewhumanitarian.org/news-feature/2024/04/08/why-cholera-killing-thousands-southern-africa>

Cholera: Koch's postulates

- Pacini 1854 - presence of 'comma shaped' bacteria in stools.
- Snow 1854 - cholera is water borne.
- Koch 1883 - isolation of *Vibrio cholerae*.
- Gradual introduction of improved sanitation in Europe & North America.

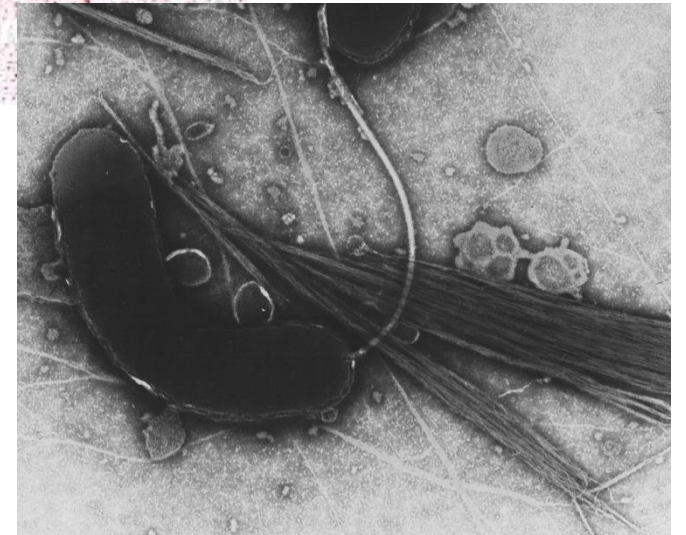
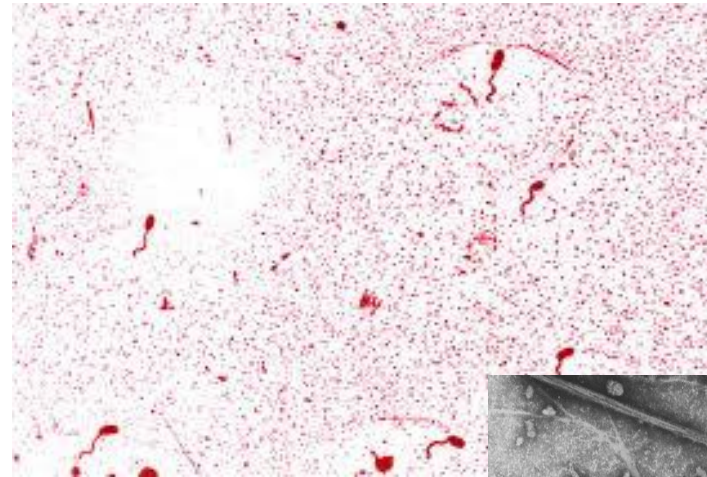


Robert Koch (third from the right) on a cholera research expedition in Egypt in 1884, one year after he identified *V. cholerae*, the microbe responsible for cholera.

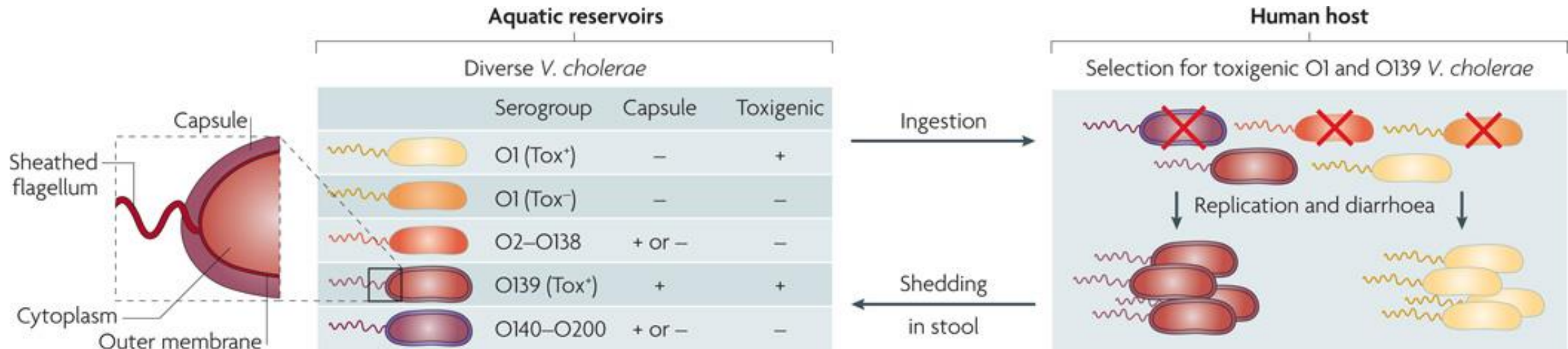
Hamlin, C. (2009). *Cholera: the biography*. Oxford, UK: Oxford University Press.

Cholera: the bacterium, *Vibrio cholerae*

- A Gram-negative bacterium, commonly found in brackish estuarine water.
- Differentiated serologically on the basis of the O antigen of its LPS.
- Cholera toxin-producing strains of the O1 and O139 serogroups cause the great majority of disease.
- All pathogenic *V. cholerae* express cholera toxin (Ctx) and toxin co-regulated pilus (Tcp).



Cholera: the genes, encoding capsule and toxin



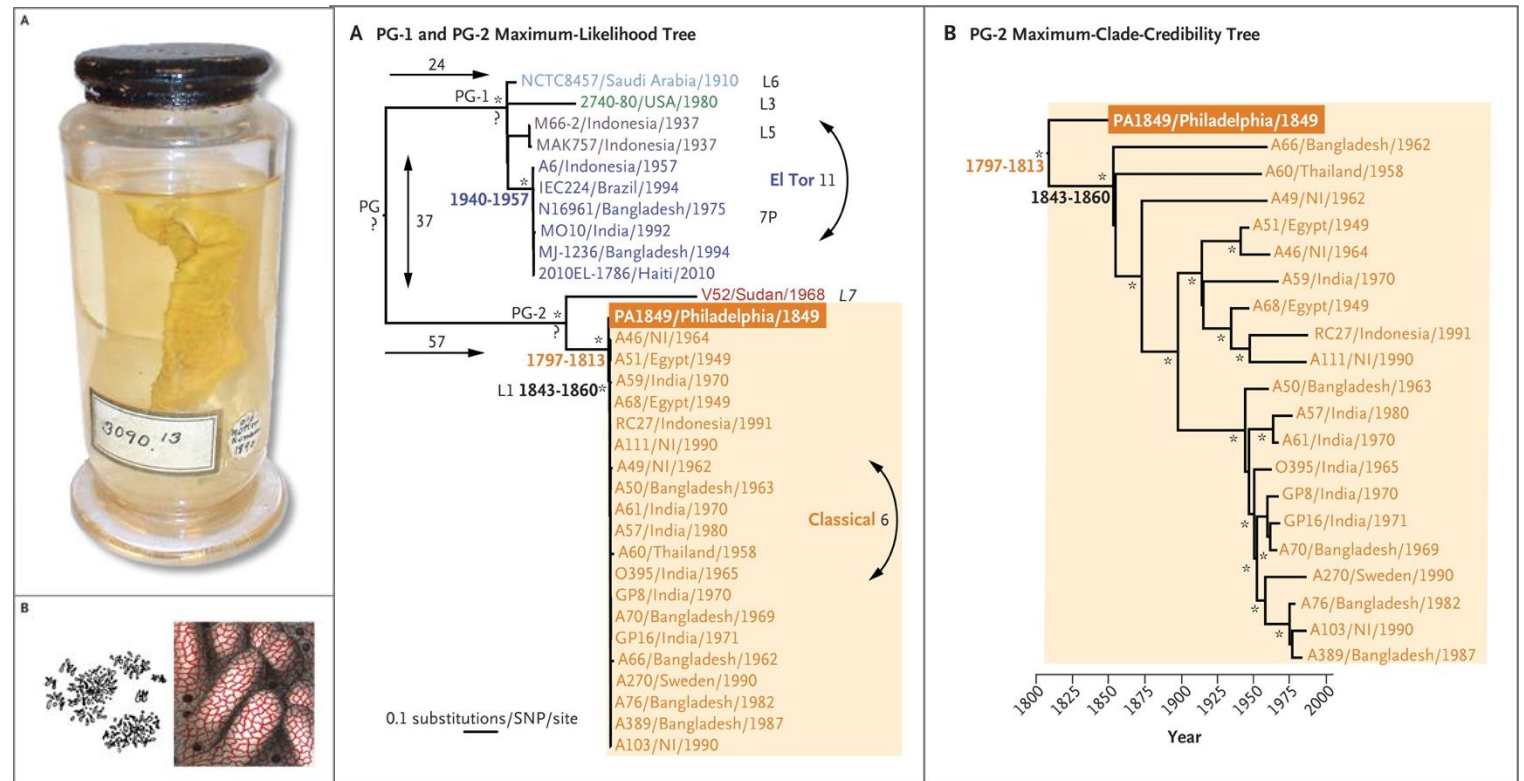
- More than 200 serogroups (O1- O200) of *Vibrio cholerae* exist.
- Only a subset of O1 and O139 are toxigenic (Tox⁺) and capable of causing cholera when ingested.
- The *tcp* gene is located on a pathogenicity island, the *ctxA* and *ctxB* genes on a filamentous phage.
- These variants are selected for in the host, Tox⁻ are selected against

Nelson E.J. et al (2009). Cholera transmission: the host, pathogen and bacteriophage dynamic. *Nature reviews* **7**, 693-702

Cholera: the power of genomics,

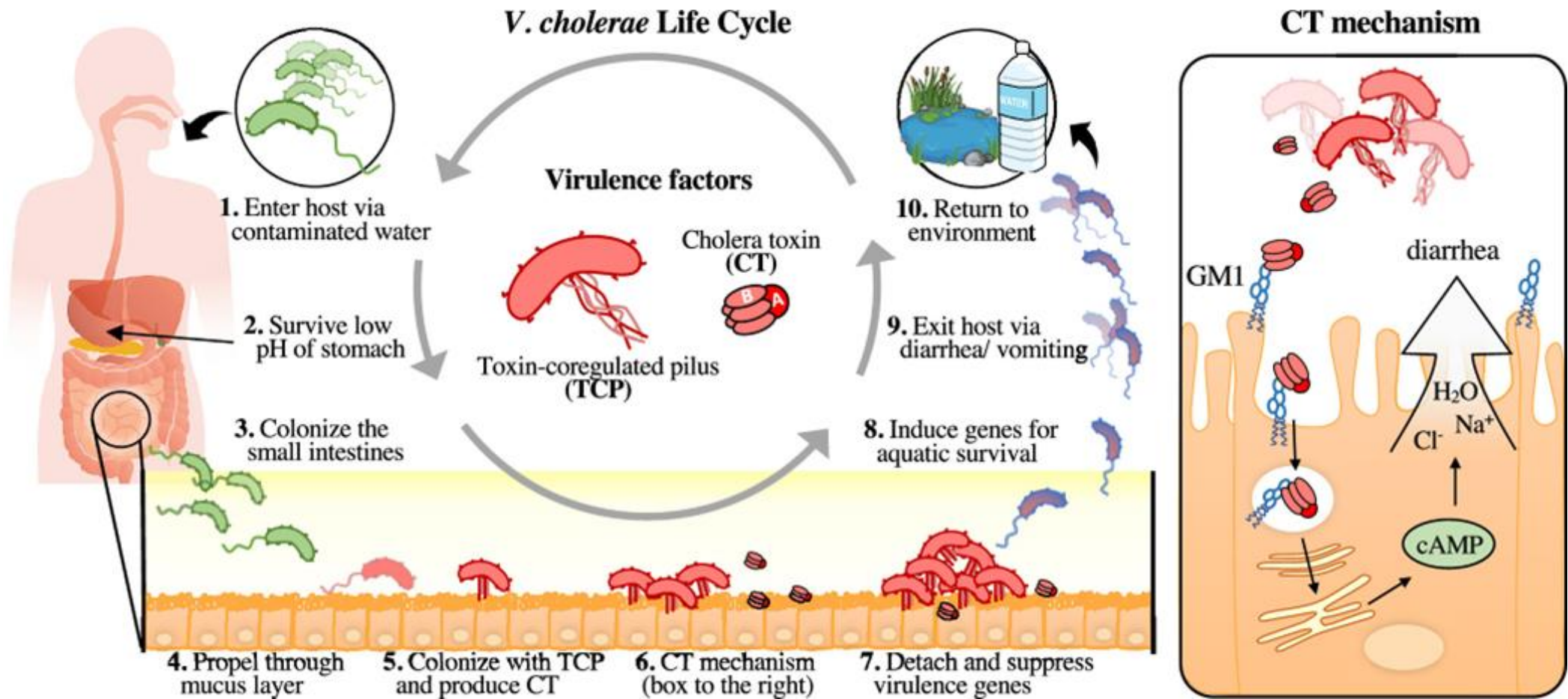
2nd Pandemic strain, Philadelphia, 1849

- A whole genome sequence was reconstructed
 - from preserved intestinal material from a cholera victim.
- The *V. cholerae* strain was O1 biotype,
 - 95-7% similarity to 'Classical' O395 genome.
- One or more tandem *ctx* phage sequences,
 - VPI-1 and VPI-2 present.



Devault, A. M., Golding, G. B., Waglechner, N., Enk, J. M., Kuch, M., Tien, J. H., Shi, M., Fisman, D. N., Dhody, A. N., Forrest, S., Bos, K. I., Earn, D. J., Holmes, E. C. & Poinar, H. N. (2014). Second-pandemic strain of *Vibrio cholerae* from the Philadelphia cholera outbreak of 1849. *N Engl J Med.* **370**, 334-340

Cholera: the dysbiosis

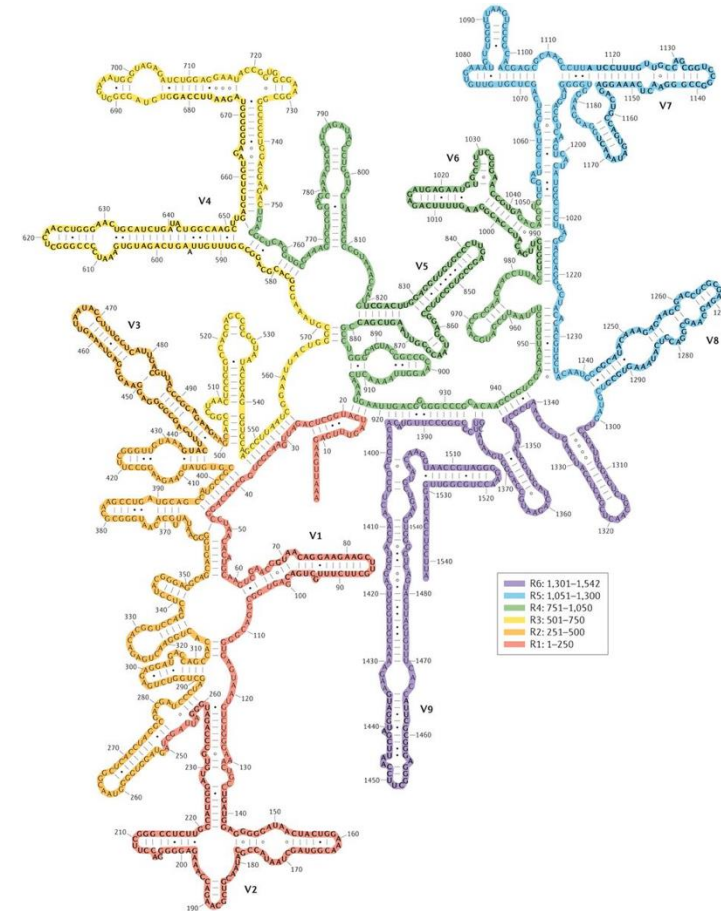


Chac, D., Dunmire, C. N., Singh, J. & Weil, A. A. (2021). Update on Environmental and Host Factors Impacting the Risk of *Vibrio cholerae* Infection. *ACS Infect Dis.* **7**, 1010-1019

Exercise, searching an NCBI database with 16SrRNA
sequences

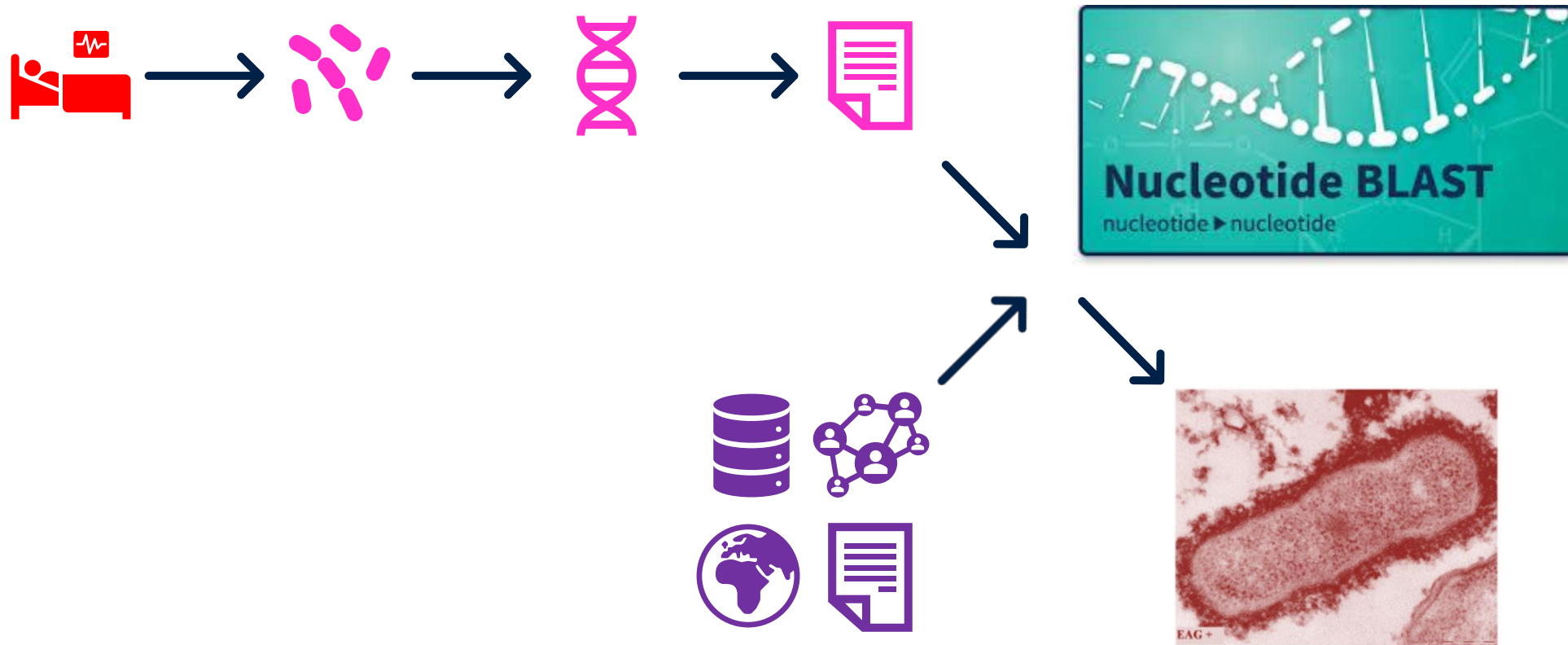
Single gene case study: 16S rRNA gene analysis

- 16S rRNA is a functionally essential structural component of the ribosome, highly conserved.
- Varies among organisms, although most variation is at the genus level.
- 16S rRNA gene, ~1.5Kbp, with:
 - universal, conserved regions, which can be used to generate primers;
 - nine variable regions which can be used as signatures to determine genus/species.



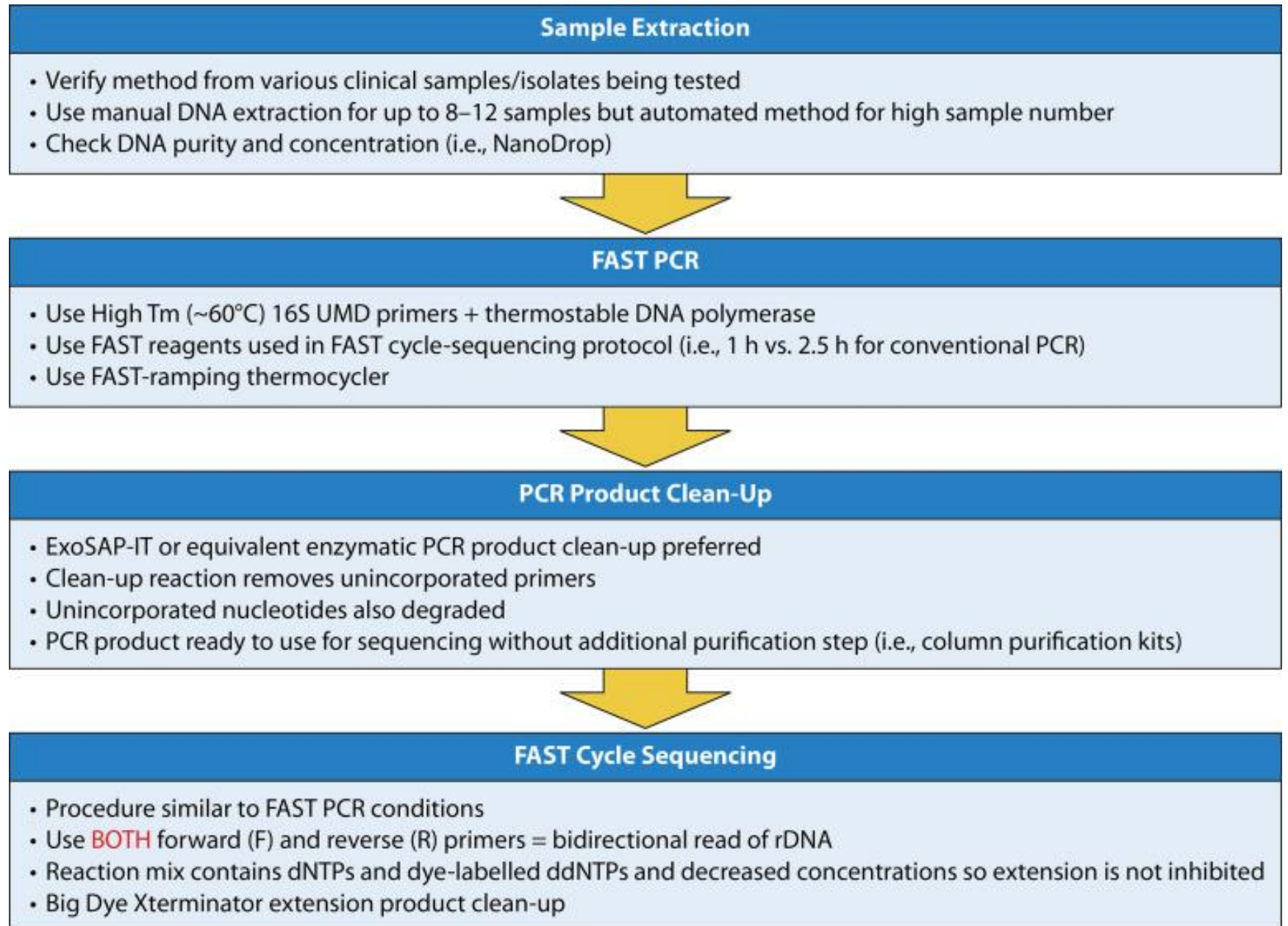
Yarza, P., Yilmaz, P., Pruesse, E., Glockner, F. O., Ludwig, W., Schleifer, K. H., Whitman, W. B., Euzéby, J., Amann, R. & Rossello-Mora, R. (2014). Uniting the classification of cultured and uncultured bacteria and archaea using 16S rRNA gene sequences. *Nat Rev Microbiol.* **12**, 635-645.

16S rRNA process overview



16S rRNA sequencing in the clinic

Church, D. L., Cerutti,
L., Gurtler, A., Griener,
T., Zelazny, A. & Emler,
S. (2020). Performance
and Application of 16S
rRNA Gene Cycle
Sequencing for Routine
Identification of
Bacteria in the Clinical
Microbiology
Laboratory. *Clin
Microbiol Rev.* **33**,
e00053-1



16s rRNA BLAST search online (1)

- Navigate to NCBI BLAST home page:
 - <https://blast.ncbi.nlm.nih.gov/Blast.cgi>

- Click on:



- Under 'Enter Query sequence' paste in a test sequence (Test 1-3)
 - https://github.com/WCSCourses/Molecular_Approaches_Clinical_Microbiology_2025/blob/main/course_data_2025/bioinformatics/16S/16S_samples.fas
 - <https://tinyurl.com/223xv5zx>
- Under 'Choose Search Set' click on 'rRNA/ITS databases'
- Check box 'Show results in a new window'. Click on

BLAST



 **Important update**
The core nucleotide database (**core_nt**) is now the default nucleotide BLAST database. [Learn more about core_nt.](#)

Standard Nucleotide BLAST

blastn blastp blastx tblastn tblastx

BLASTN programs search nucleotide databases using a nucleotide query, more...

[Reset page](#) [Bookmark](#)

Enter Query Sequence

Enter accession number(s), gi(s), or FASTA sequence(s) 

>Test_1
AATTGAAGAGTTTGCATGGCTCAGATTGAACGCTGGCGGCAGGCCTAACA
CATGCAAG
TCGAACGGTAACAGGAAGAAGCTTGCTCTTTGCTGACGAGTGGCGGACGG

Query subrange 

From

To

Or, upload file No file selected. 

Job Title

Enter a descriptive title for your BLAST search 

☐ Align two or more sequences 

Choose Search Set

Database ☐ Standard databases (nr etc.): ☒ rRNA/ITS databases ☐ Genomic + transcript databases ☐ Betacoronavirus ☐ Experimental databases

☒ 16S ribosomal RNA sequences (Bacteria and Archaea)  [Targeted Loci Project Information](#)

Organism Optional ☐ exclude
Enter organism common name, binomial, or tax id. Only 20 top taxa will be shown 

Exclude Optional ☐ Models (XM/XP) ☐ Uncultured/environmental sample sequences

Limit to Optional ☐ Sequences from type material

Entrez Query Optional  [Create custom database](#)
Enter an Entrez query to limit search 

Program Selection

Optimize for ☒ Highly similar sequences (megablast) 
☐ More dissimilar sequences (discontiguous megablast)
☐ Somewhat similar sequences (blastn)
Choose a BLAST algorithm 

BLAST Search database 16S ribosomal RNA sequences (Bacteria and Archaea) using Megablast (Optimize for highly similar sequences)

☒ Show results in a new window

Note: Parameter values that differ from the default are highlighted in yellow and marked with  sign

+ Algorithm parameters

What bacterium is it?

- Look at the results of the search (scroll down the page as necessary).
- What is the likely organism that the 16S rRNA sequence came from?

Sample	Candidate Bacterium
Test 1	
Test 2	
Test 3	

Encapsulated Theme: applications of molecular techniques

Molecular Approaches to Clinical Microbiology in Africa 2025

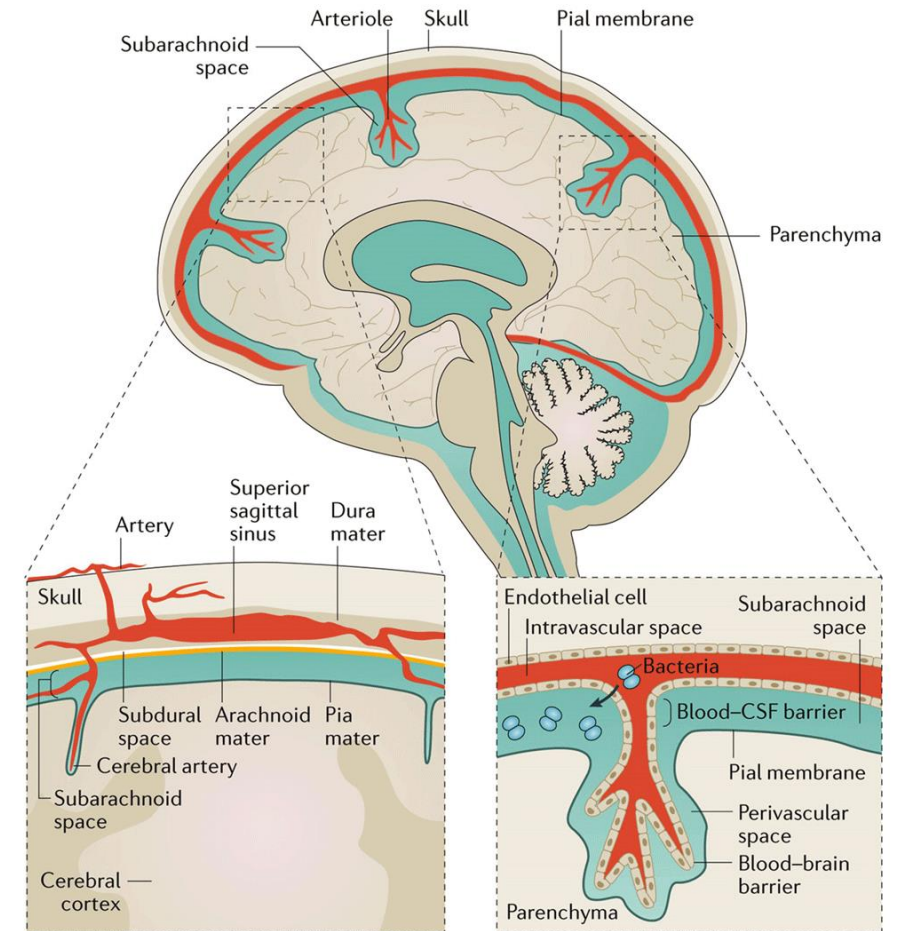
Martin Maiden, Keith Jolley



UNIVERSITY OF
OXFORD

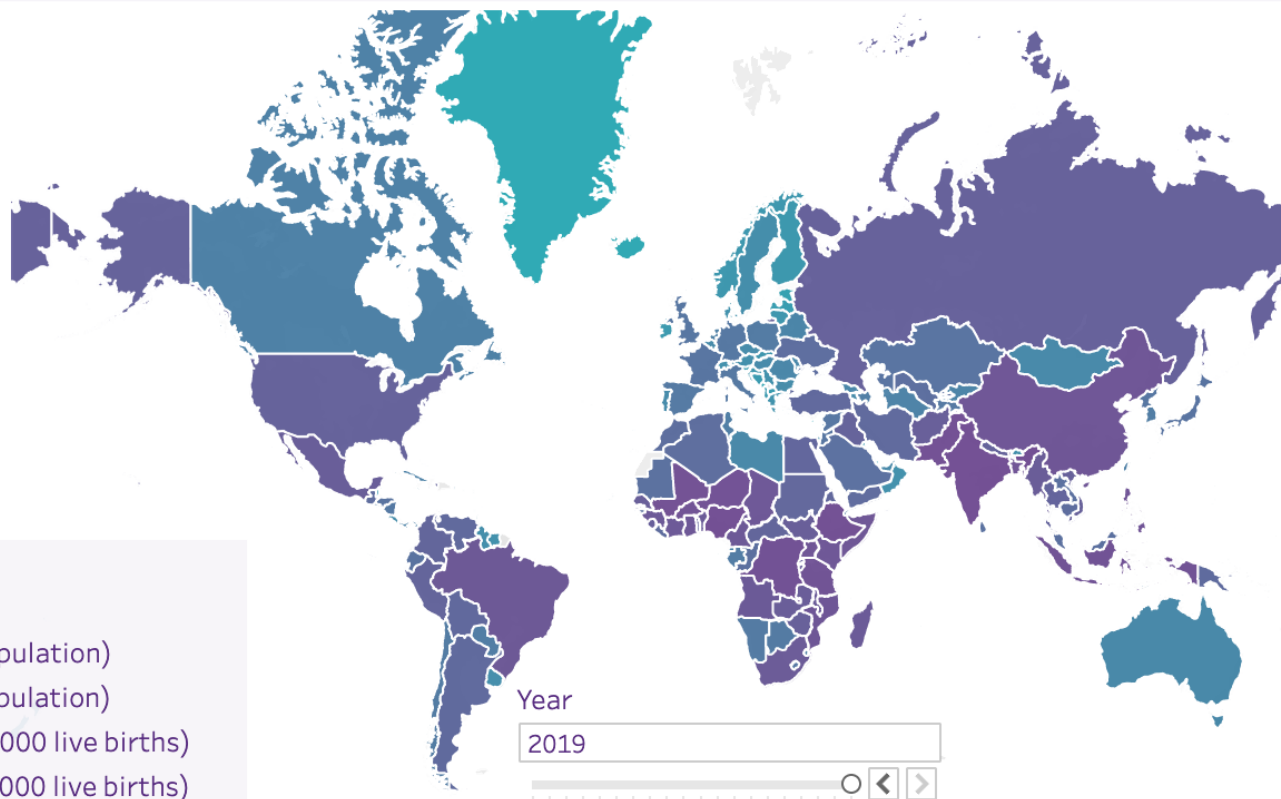
Meningitis

- Meningitis:
 - Inflammation of the meninges (tissues around the brain).
- Caused by invasion of the meninges by a pathogen:
 - Bacterial, viral, fungal, parasite;
 - Can be accompanied by other pathologies e.g. septicaemia (blood poisoning).
- Severe and frequently fatal especially when caused by bacteria:
 - Survivors frequently suffer sequelae e.g. digit or limb loss, brain damage, deafness.
- Diagnosis difficult
 - but essential for swift appropriate treatment.



Rodrigues, C. M. C. & Maiden, M. C. J. (2018). A world without bacterial meningitis: how genomic epidemiology can inform vaccination strategy. *F1000Res* **7**, 401.

Meningitis and Neonatal Sepsis



N.B IHME produce estimates for all ages, causes and years. Case estimates broken down by cause are only available for the years: 2000, 2005, 2010, 2015, 2017 and 2019

Rank of country by Deaths
(highest to lowest)

1 203

Age Range

(AII)

Disease Category

(All)

Cause

(All)

Meningitis Belt ?

(All)

WHO Region

(All)

Income Level

(All)

Source of Estimates

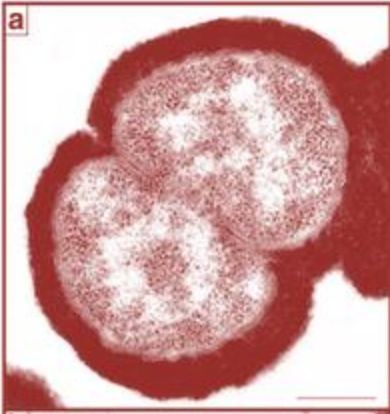
IHME

Reset

Estimated Total Deaths 462,452

Estimated Total Cases 8,427,054

Meningitis causing organisms

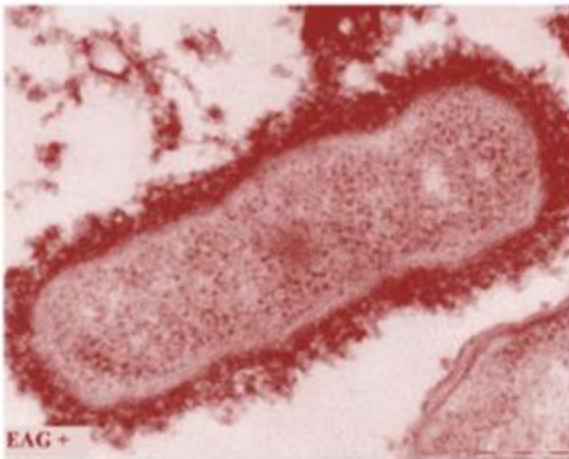
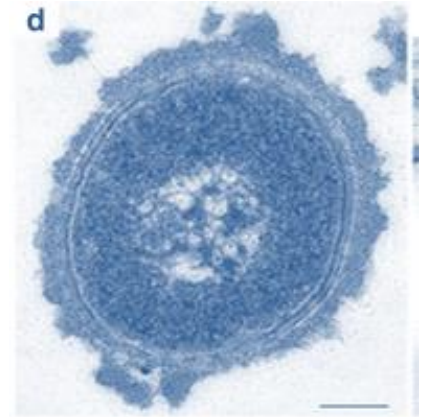


Neisseria meningitidis
Meningococcus
12 capsular serogroups
Genome: 2.27Mbp

Ganesh, K. et al. (2017) Molecular characterization of invasive capsule null *Neisseria meningitidis* in South Africa. *BMC Microbiol.* **17**(1), 40.

Streptococcus pneumoniae
Pneumococcus
>100 capsular serotypes
Genome: 2.04Mbp

Ndlangisa, K.M., et al. (2016) Two cases of serotypeable and non-serotypeable variants of *Streptococcus pneumoniae* detected simultaneously during invasive disease. *BMC Microbiol.* **16**, 126.

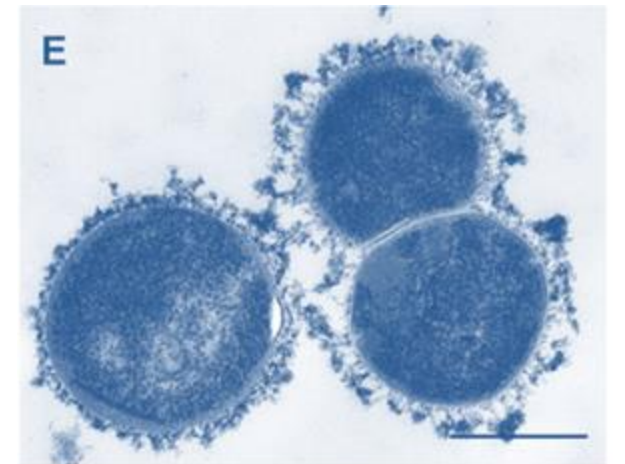


Haemophilus influenzae
'Hib'
6 capsular serotypes
Genome: 1.83Mbp

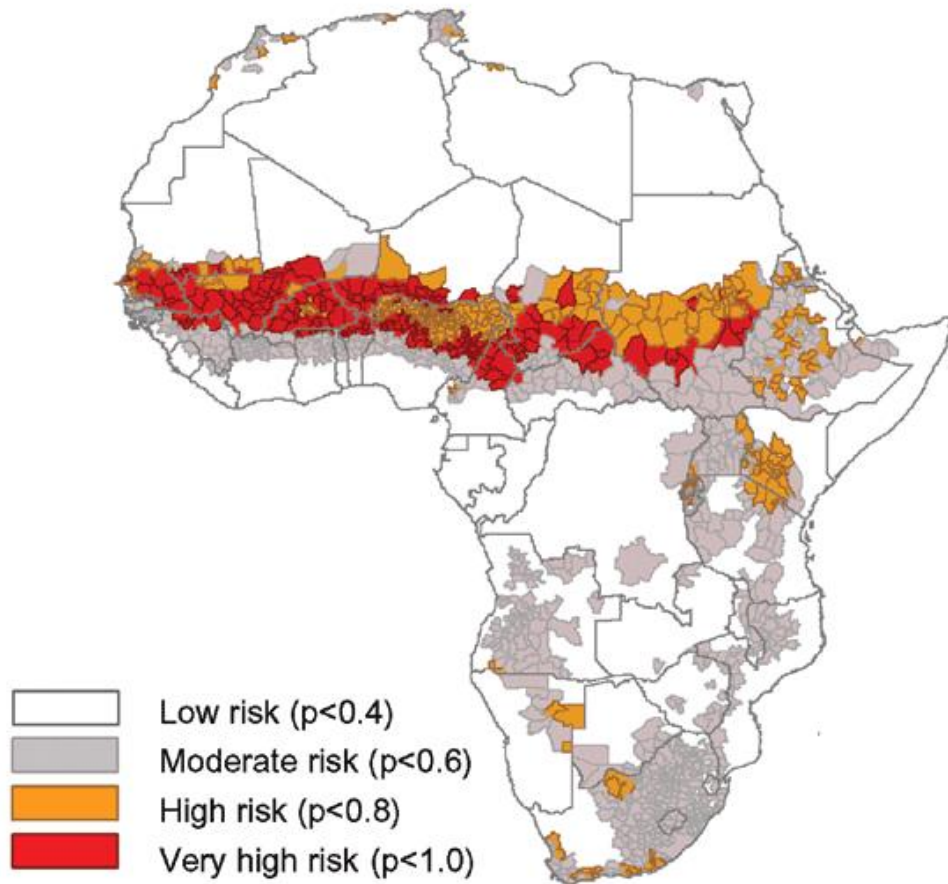
Schouls L., et al. (2008) Two variants among *Haemophilus influenzae* serotype b strains with distinct *bcs4*, *hcsA* and *hcsB* genes display differences in expression of the polysaccharide capsule. *BMC Microbiol.* **8**, 35.

Streptococcus agalactiae
Group B Strep. (GBS)
10 capsular serotypes
Genome: 2.16Mbp

Lecours M.P., et al. (2012) Sialylation of *Streptococcus suis* serotype 2 is essential for capsule expression but is not responsible for the main capsular epitope. *Microbes Infect.* **14**(11), 941-50.



The meningitis belt



Meningitis belt extends from Ethiopia to Senegal.

Sudan, Ethiopia, Chad, Niger, Northern Nigeria, Burkina Faso, and Mali are considered hyper-endemic.

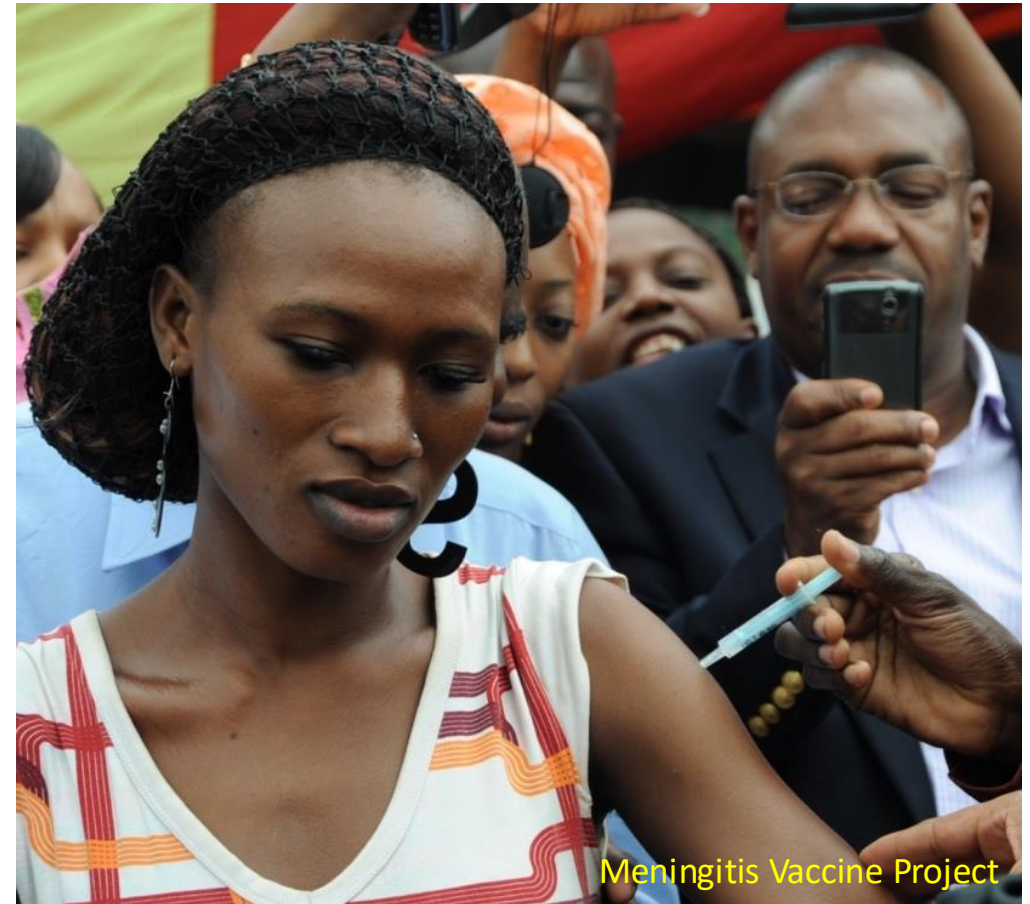
- **1905**, first documented epidemic, Northern Nigeria.
- **1919-1924**, second cycle, >45,000 deaths, Northern Nigeria.
- **1935-1937**, third cycle, Nigeria, 6,456 deaths.
- 1951-60, 340,000 cases, 53,000 deaths.
- 1996-1997, 300,000 cases, 30,000 deaths

Molesworth, A. M., Cuevas, L. E., Connor, S. J., Morse, A. P. & Thomson, M. C. (2003). Environmental risk and meningitis epidemics in Africa. *Emerg Infect Dis* **9**, 1287-1293.

Greenwood, B. (1999). Manson Lecture. Meningococcal meningitis in Africa. *Trans R Soc Trop Med Hyg* **93**, 341-353

MenAfriVac®: combatting epidemic meningococcal disease in Africa

- Meningitis Vaccine Project (MVP):
 - Gates funded, partnership with WHO and PATH;
 - Innovative development – Northern technology, southern manufacturer;
 - Low cost (less than \$1 per dose, for sustainable use;
- MenAfriVac®, a polysaccharide-protein conjugate vaccine, introduced 2010 in Burkina Faso;
 - Everyone aged 1-29 years immunised;
 - Immediate reduction in disease levels.



Diomande, F. V. K., Djingarey, M. H., Daugla, D. M., Novak, R. T., Kristiansen, P. A., Collard, J. M., Gamougam, K., Kandolo, D., Mbakuliyemo, N., Mayer, L., Stuart, J., Clark, T., Tevi-Benissan, C., Perea, W. A., Preziosi, M. P., LaForce, F. M., Caugant, D., Messonier, N., Walker, O. & Greenwood, B. (2015). Public Health Impact After the Introduction of PsA-TT: The First 4 Years. *Clinical Infectious Diseases* 61, S467-S472.

Global

Regions ▾



Select language ▾



Health Topics ▾

Countries ▾

Newsroom ▾

Emergencies ▾

Data ▾

About WHO ▾

Home / Initiatives / Defeating Meningitis by 2030



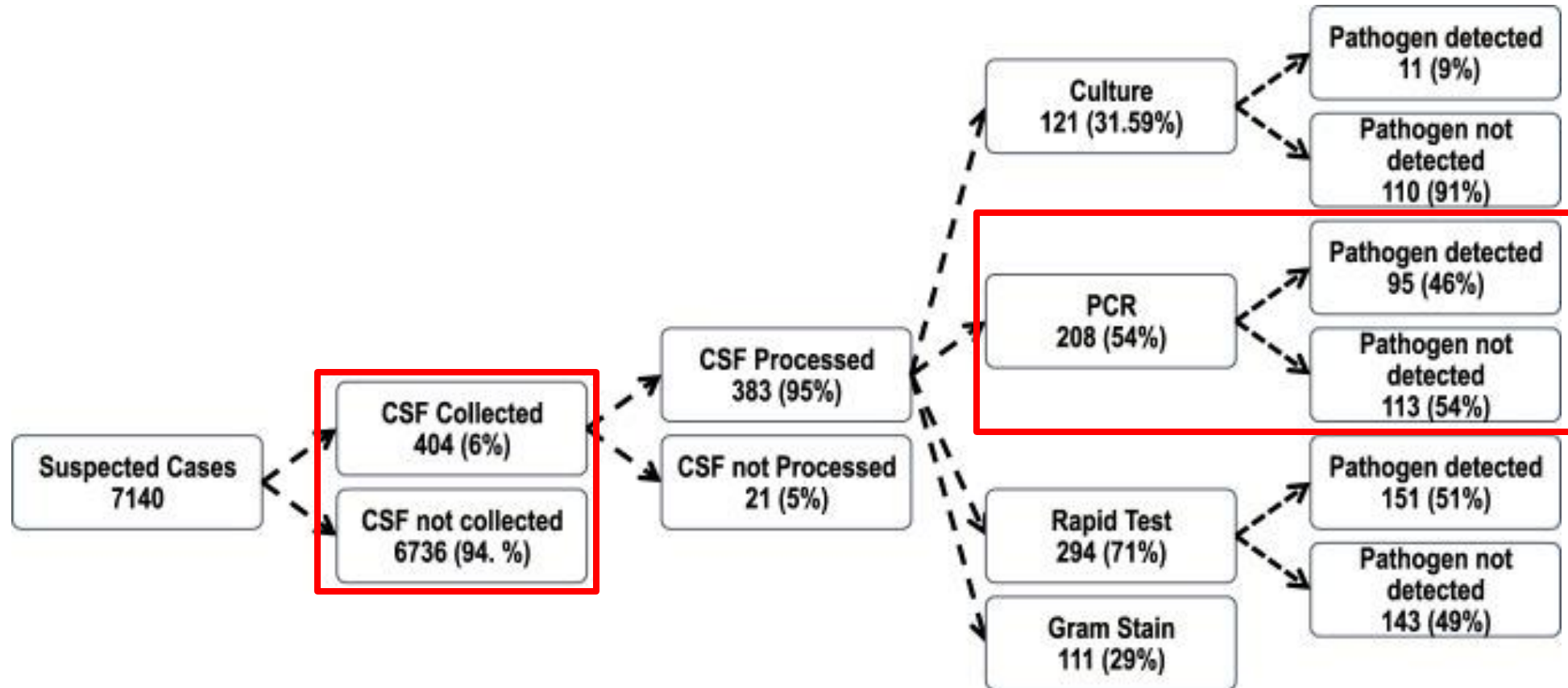
<https://www.who.int/initiatives/defeating-meningitis-by-2030>

Global road map priorities for response, care and prevention

- Achievement of high immunization coverage, development of new affordable vaccines, and improved prevention strategies and outbreak response;
 - Speedy diagnosis and optimal treatment for patients;
 - Good data to guide prevention and control efforts;
- Care and support for those affected, focusing on early recognition and improved access to care and support for after-effects; and
 - Advocacy and engagement, to ensure high awareness of meningitis, accountability for national plans, and affirmation of the right to prevention, care and after-care services.

<https://www.who.int/news-room/events/detail/2024/04/26/default-calendar/the-first-high-level-meeting-to-defeat-meningitis---institut-pasteur-paris--france>

Increasing importance of molecular diagnosis and isolate characterisation in Africa

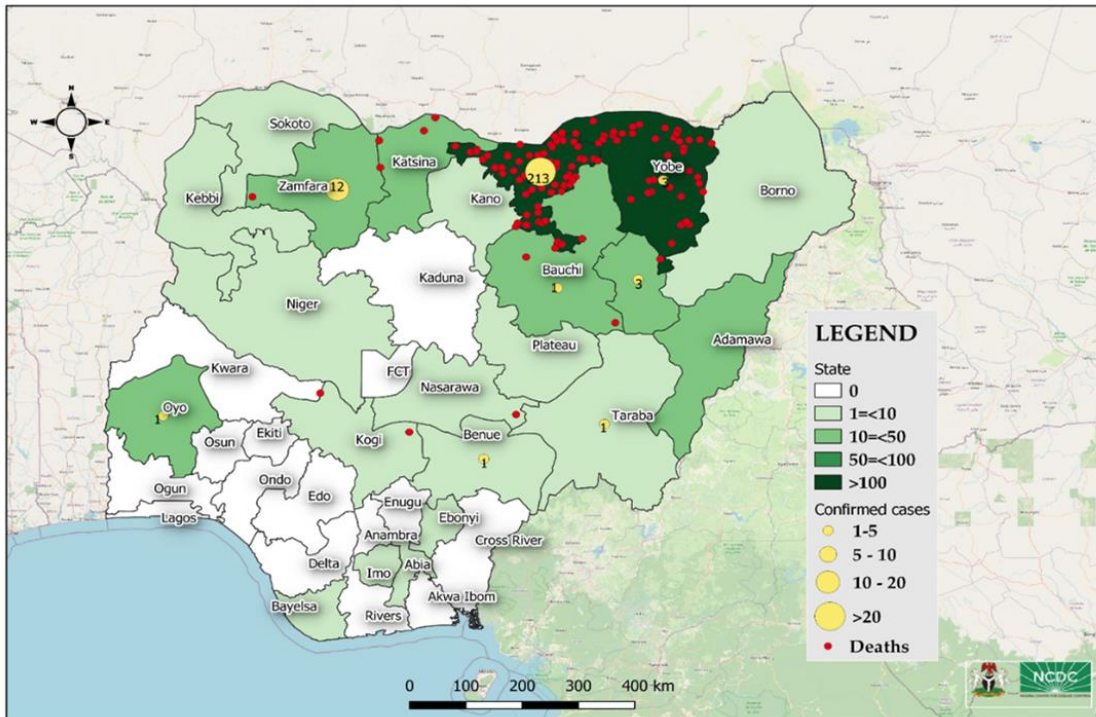


Kwambana-Adams, B. A., Amaza, R. C., Okoi, C., Rabi, M., Worwui, A., Foster-Nyarko, E., Ebruke, B., Sesay, A. K., Senghore, M., Umar, A. S., Usman, R., Atiku, A., Abdullahi, G., Buhari, Y., Sani, R., Bako, H. U., Abdullahi, B., Yarima, A. I., Sikiru, B., Moses, A. O., Popoola, M. O., Ekeng, E., Olayinka, A., Mba, N., Kankia, A., Mamadu, I. N., Okudo, I., Stephen, M., Ronveaux, O., Busuttil, J., Mwenda, J. M., Abdulaziz, M., Gummi, S. A., Adedeji, A., Bitu, A., Omar, L., Djingarey, M. H., Alemu, W., D'Alessandro, U., Ihekweazu, C. & Antonio, M. (2018). Meningococcus serogroup C clonal complex ST-10217 outbreak in Zamfara State, Northern Nigeria. *Scientific reports*. 8, 14194.

Meningitis in Nigeria 2022-2024

1686 suspected cases 1 October 2022 - 16 April 2023

<https://www.who.int/emergencies/disease-outbreak-news/item/2023-DON454>



5 in 1 meningitis vaccine introduced

<https://www.bbc.co.uk/programmes/w3ct5t8d>

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Nigeria rolls out world's first 5-in-1 meningitis vaccine

World first 5-in-1 meningitis vaccine rolled out in Nigeria; How Brazil is coping with long Covid; Using patches of grafted skin to detect donated organ rejection

Available now
26 minutes

Show more
Download

Ashinze, P., Mafua, N. & Obafemi, E.(2024). Nigeria rolls out novel meningitis vaccine. *Lancet*. 403, 2373.

Exercise, searching an NCBI database with *Neisseria*
16S rRNA sequences

16s rRNA BLAST search online (2)

- Navigate to NCBI BLAST home page:
 - <https://blast.ncbi.nlm.nih.gov/Blast.cgi>

- Click on:



- Under 'Enter Query sequence' paste in a test sequence (Test 4-6)
 - https://github.com/WCSCourses/Molecular_Approaches_Clinical_Microbiology_2025/blob/main/course_data_2025/bioinformatics/16S/16S_samples.fas
 - <https://tinyurl.com/223xv5zx>
- Under 'Choose Search Set' click on 'rRNA/ITS databases'
- Check box 'Show results in a new window'. Click on

BLAST



Important update
The core nucleotide database (**core_nt**) is now the default nucleotide BLAST database. [Learn more about core_nt.](#)

Standard Nucleotide BLAST

BLASTN programs search nucleotide databases using a nucleotide query, more...

[Reset page](#) [Bookmark](#)

Enter Query Sequence

Enter accession number(s), gi(s), or FASTA sequence(s) 

```
>Test_1
AATTGAAGAGTTTGATCATGGCTCAGATTGAACGCTGGCGGCAGGCCTAACA
CATGCAAG
TCGAACGGTAACAGGAAGAAGCTTGCTCTTTGCTGACGAGTGGCGGACGG
```

Query subrange 

From

To

Or, upload file No file selected. 

Job Title

Enter a descriptive title for your BLAST search 

☐ Align two or more sequences 

Choose Search Set

Database ☐ Standard databases (nr etc.): ☒ rRNA/ITS databases ☐ Genomic + transcript databases ☐ Betacoronavirus ☐ Experimental databases

16S ribosomal RNA sequences (Bacteria and Archaea) 

[Targeted Loci Project Information](#)

Organism Optional ☐ exclude [Add organism](#)

Enter organism common name, binomial, or tax id. Only 20 top taxa will be shown 

Exclude Optional ☐ Models (XM/XP) ☐ Uncultured/environmental sample sequences

Limit to Optional ☐ Sequences from type material

Entrez Query Optional [YouTube](#) [Create custom database](#)
Enter an Entrez query to limit search 

Program Selection

Optimize for ☒ Highly similar sequences (megablast)

☐ More dissimilar sequences (discontiguous megablast)

☐ Somewhat similar sequences (blastn)

Choose a BLAST algorithm 

BLAST Search database 16S ribosomal RNA sequences (Bacteria and Archaea) using Megablast (Optimize for highly similar sequences)

☒ Show results in a new window

Note: Parameter values that differ from the default are highlighted in yellow and marked with  sign

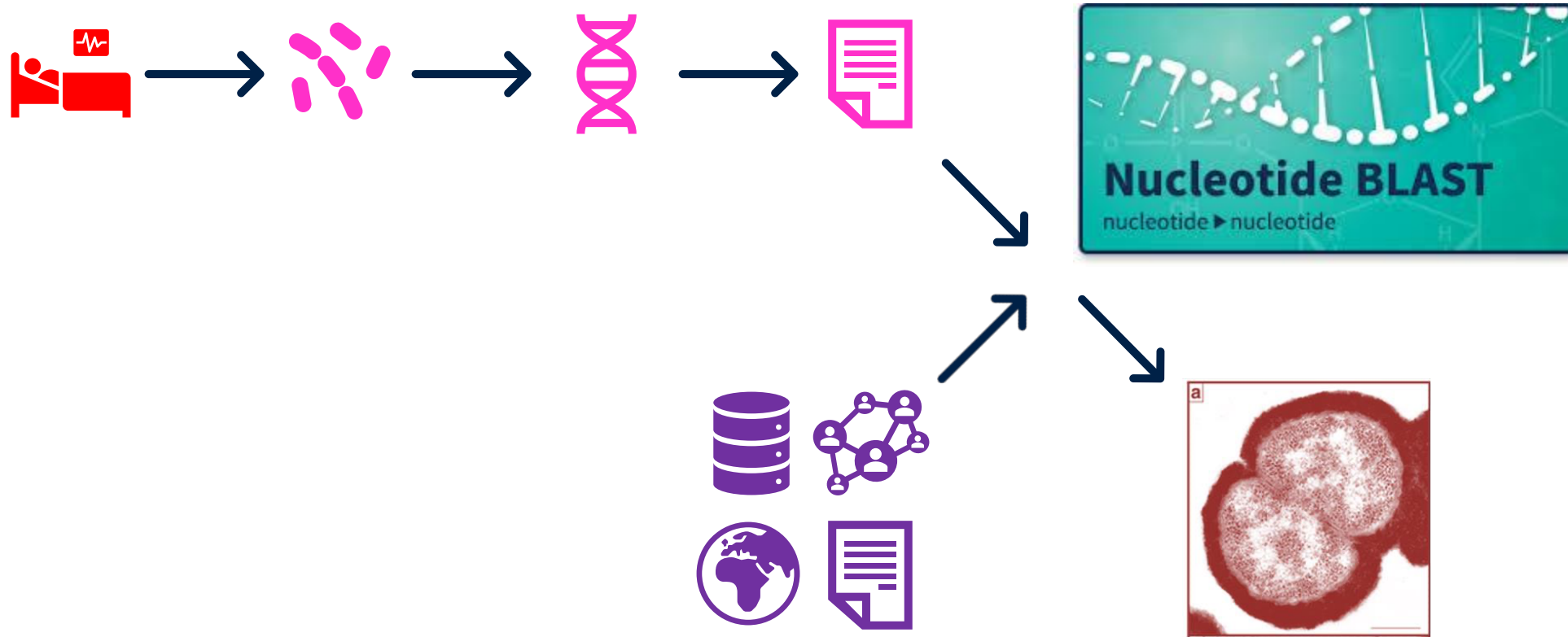
+ Algorithm parameters

What bacterium is it?

- Look at the results of the search (scroll down the page as necessary).
- What is the likely organism that the 16S rRNA sequence came from?

Sample	Candidate Bacterium
Test 4	
Test 5	
Test 6	

16S rRNA summary



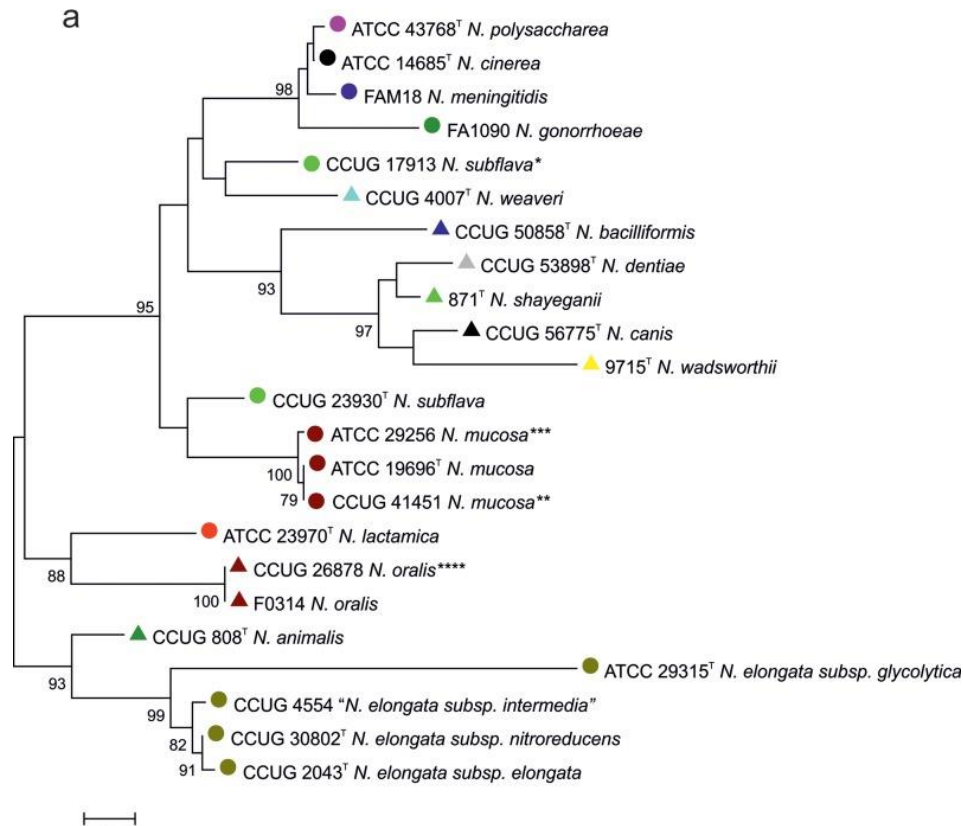
Case study: *rplF* assay

- **Problem:** rapid cost effective determination of *Neisseria* species from 1000s of isolates obtained in the MenAfriCar surveys.
 - 16s rRNA sequencing too cumbersome and insufficient discrimination or resolution.
- **Solution:** identify a short sequenceable gene fragment (~400bp in length) diagnostic for species.
- **Implementation:** Phylogeny of *Neisseria* species generated and compared to phylogenies of individual genes
 - A fragment of *rplF* gene was congruent with clusters in the phylogeny of all genes and used for the assay.

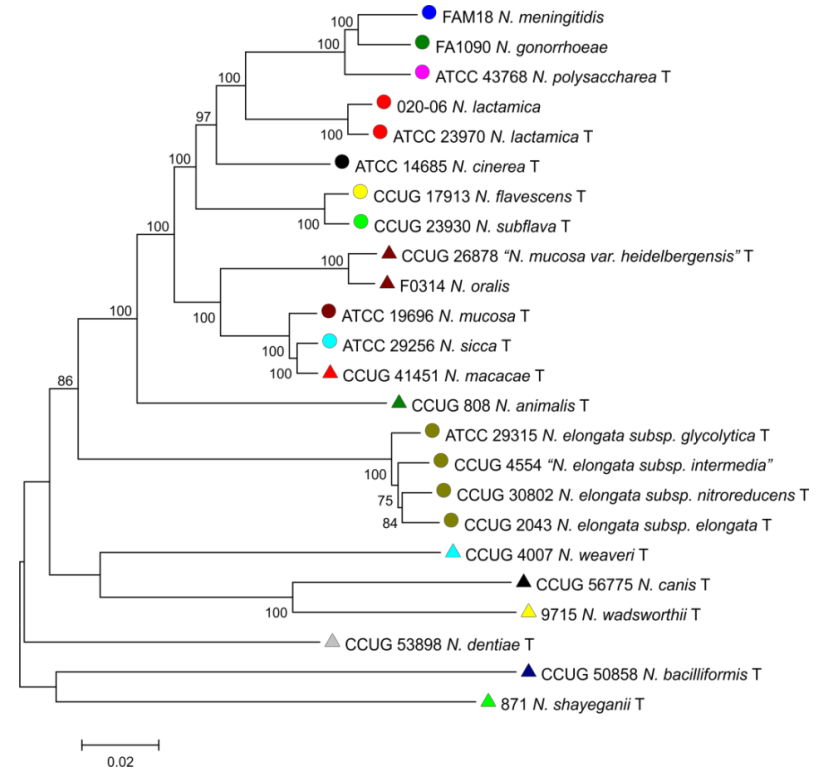
Bennett, J. S., Watkins, E. R., Jolley, K. A., Harrison, O. B. & Maiden, M. C. (2014). Identifying *Neisseria* species using the 50S ribosomal protein L6 (*rplF*) gene. *J Clin Microbiol* **52**, 1375-1381.

Comparison of 16s rRNA sequencing and *rplF* sequencing

16s rRNA gene fragment phylogeny (single rRNA encoding gene)



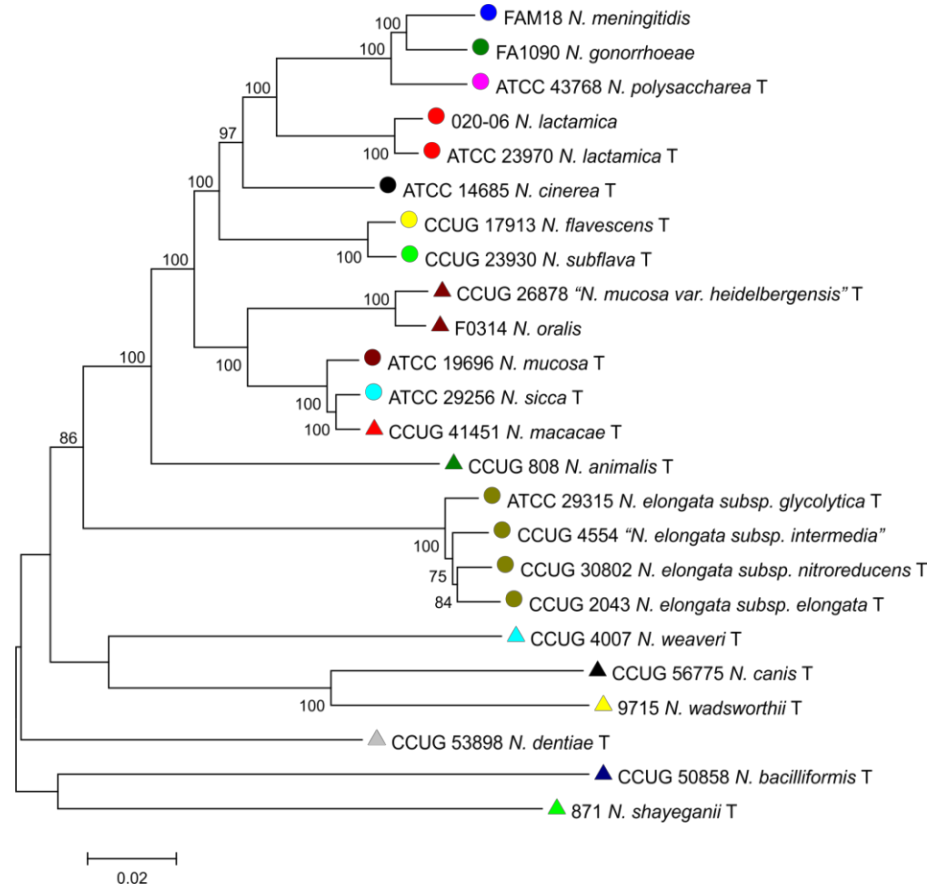
rMLST phylogeny (53 ribosomal protein encoding genes)



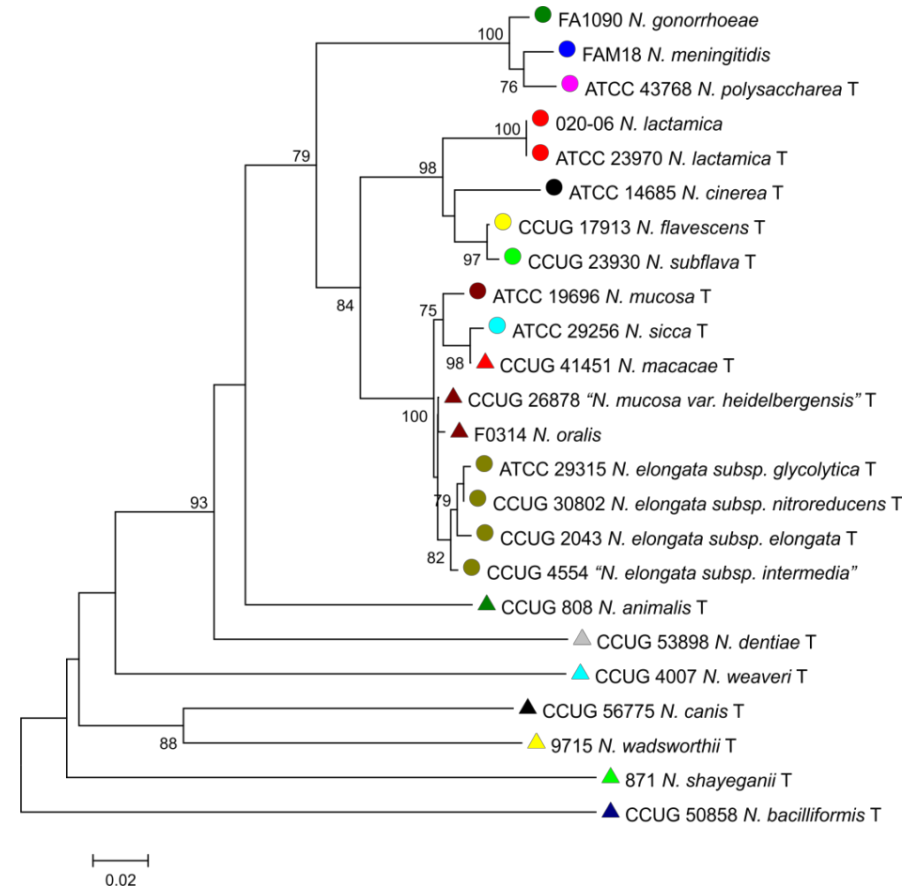
Bennett, J. S., Watkins, E. R., Jolley, K. A., Harrison, O. B. & Maiden, M. C. (2014). Identifying *Neisseria* species using the 50S ribosomal protein L6 (*rplF*) gene. *J Clin Microbiol* 52, 1375-1381.

Rapid species assignment: *rplF* sequence

rMLST phylogeny (53 genes)

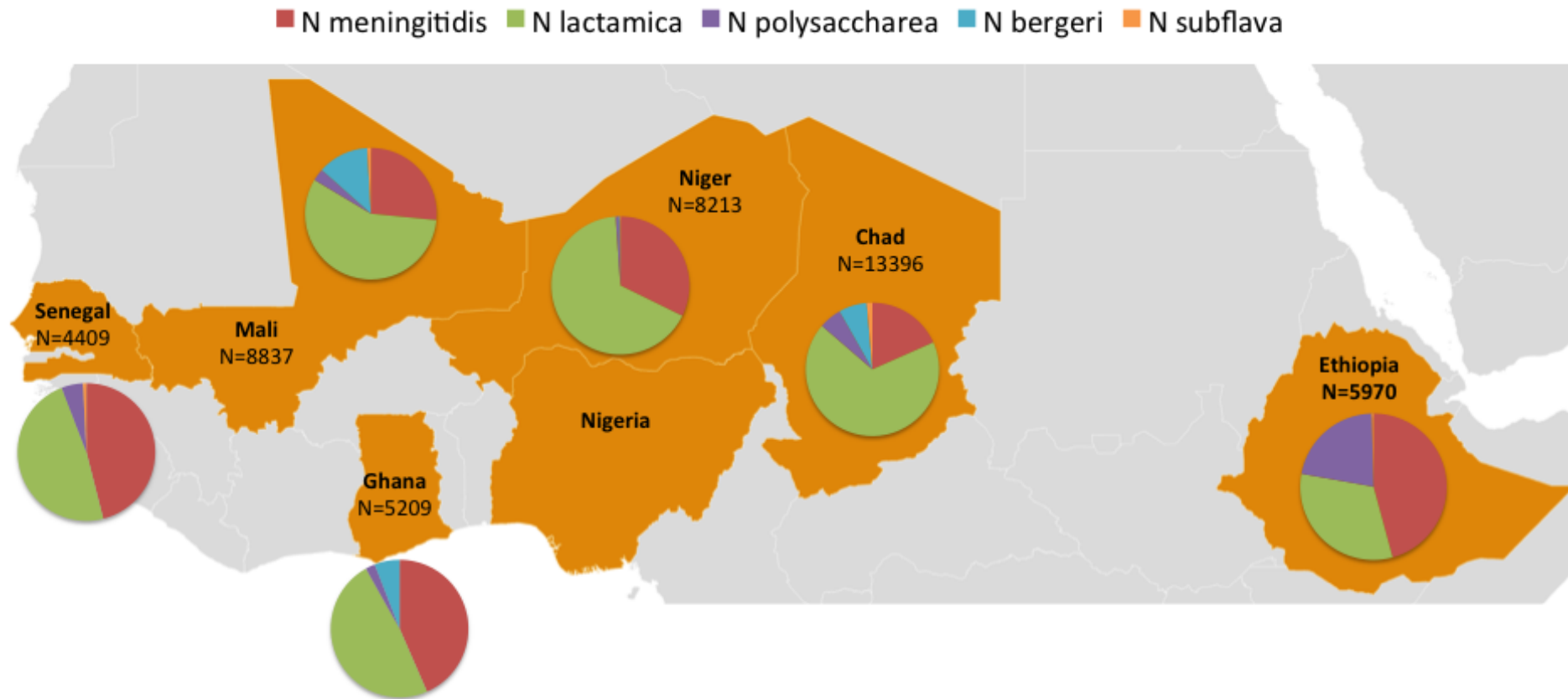


rplF fragment (413bp) phylogeny



Bennett, J. S., Watkins, E. R., Jolley, K. A., Harrison, O. B. & Maiden, M. C. (2014). Identifying *Neisseria* species using the 50S ribosomal protein L6 (*rplF*) gene. *J Clin Microbiol* 52, 1375-1381.

Neisseria species distribution



Diallo, K., *et al.*, Greenwood, B. M. & Maiden, M. C. (2016). Pharyngeal carriage of *Neisseria* species in the African meningitis belt. *J Infect* **72**, 667-677.